Entropy Collapse Has a Direction: The Missing Visitation Channel in Token-Level Entropy Control for RLVR

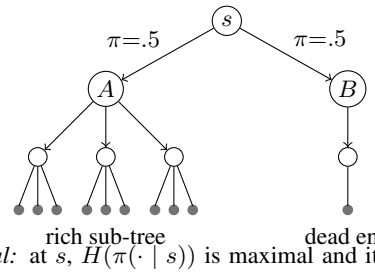
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Abstract

Reinforcement learning with verifiable rewards (RLVR) drives a policy's entropy toward zero, and a recent family of methods counters this by re-weighting the policy loss with a per-token estimate of each update's entropy cost. These methods account for how visited states sharpen, but not for how an update changes *which states are visited*. We show that the gradient of the trajectory entropy splits exactly into a *local* channel — the change of the conditional distribution at visited states — and a *visitation* channel — the change in the probability of reaching them — and that token-level estimators capture only the first. The local estimate vanishes, correctly to first order, at a flat branch point, which is exactly where the visitation prefactor is largest, so no refinement of the local estimator recovers the missing term. We derive the visitation term as a per-token REINFORCE advantage A_H whose reward-to-go is the remaining conditional entropy, made estimable by a sibling-prefix baseline that a fixed tree-structured rollout supplies. A_H sums to zero within each sibling set, so it can change trajectory diversity without changing aggregate token entropy. A five-arm experiment over three paired seeds observes this dissociation: changing the damping mechanism shifts the aggregate entropy by $-.0419$ and $+.0184$, whereas changing only the branch-specific A_H signal shifts it by $-.0077$ and $+.0044$. We do not establish an accuracy gain from the correction — a pre-registered accuracy test is not supported — and the experiments instead support the structural claim that aggregate entropy cannot diagnose the visitation channel.

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model's output distribution. Measured as the mean per-token entropy, the sharpening is fast, monotone and, past a point, coincident with the end of useful learning. The standard remedies spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose most entropy, or, most recently, a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token's update destroys (Hao et al., 2026).



Local: at s, $H(\pi(\cdot \mid s))$ is maximal and its first-order change is 0.
Visitation: moving mass from A to B removes A's downstream diversity.

Figure 1: A branch can be locally maximally entropic while a small change in which branch is visited destroys most of the downstream trajectory diversity. Token-level entropy control sees only the first fact.

These remedies account for how the conditional distribution at a visited state sharpens. They do not account for how an update changes which states are visited, and this omission is structural rather than a matter of estimator quality. Figure 1 shows the case that makes it concrete. At a state with two equiprobable continuations the conditional distribution is as entropic as it can be, and its first-order entropy change under a small push is zero. Yet a small shift of mass from a continuation that leads into a rich sub-tree to one that leads into a dead end removes all the downstream diversity the first carried. The branch itself is locally safe; which branch is visited is not.

We make this precise. Write $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution over whole responses. Its gradient splits, by the product rule and with no approximation, into a *local* term that holds visited states fixed — what a per-token estimate measures

— and a *visitation* term that holds per-state entropies fixed and differentiates the probability of reaching each state. The local estimate is proportional to the deviation of the sampled token's surprisal from its mean; it vanishes at a flat branch point, correctly, since the entropy loss there is second order. The visitation term's prefactor $1 - \pi_\theta(y_t \mid s_t)$ is largest at those same points.

The missing term is estimable. Causality turns it into a REINFORCE estimator whose reward-to-go is the remaining conditional entropy H_{togo}, and a baseline from rollouts that share a prefix turns that into a per-token advantage A_H. Plain group rollouts almost never share prefixes, so we schedule the sharing with a fixed tree, which we use only as a device for the estimator. Because A_H is a deviation from a sibling mean it sums to zero over each sibling set: it changes which sibling is damped, not how much damping there is on average. This gives the experiment its target. The aggregate entropy curve — the evidence usually offered for entropy control — should move when the damping mechanism changes and should not move when only the branch-specific A_H signal changes, a prediction that can fail.

Our contributions are:

- **An exact decomposition** of the trajectory-entropy gradient into a local term, which changes conditional entropy at visited states, and a visitation term, which changes which states the policy reaches (Section 3). The identity is standard in maximum-entropy RL; what is new is reading token-level entropy control through it.
- **A structural mismatch:** the local estimator used by current token-level methods vanishes at flat branch points, precisely where the visitation prefactor is largest, so the omission cannot be repaired within the local channel (Section 3.1).
- **An estimable visitation term:** a per-token REINFORCE advantage A_H whose reward-to-go is the remaining conditional entropy, with a sibling-prefix baseline, a ledger of every approximation and the sign of its error, and a bound on how many positions can carry it under any sampler (Section 4).
- **A diagnostic consequence:** because the correction is mean-preserving within sibling sets, aggregate token entropy is not an appropriate diagnostic for whether it acts (Section 4).
- **An experimental test of these predictions:** five arms that separate changes to the damping mechanism from changes to the branch-specific signal. The predicted dissociation is observed — aggregate entropy tracks the mechanism, not the signal; the second prediction, a wider spread across siblings, is not measured here (Sections 5–6.1).

We do not establish an accuracy gain from the correction: the pre-registered test of the A_H signal against a signal-free control is not supported (Section 6.3).

2 Background and Related Work

Notation. A prompt is x, a response $y = y_{1:T}$, and $s_t = (x, y_{<t})$ the state before token t. $H(\pi_\theta(\cdot \mid s_t))$ is a per-position conditional entropy and $\mathcal{H}(\pi_\theta \mid x)$ the entropy over whole responses. The surprisal of a token is $I_a := -\log \pi_\theta(a \mid s_t)$, so $H = \mathbb{E}_a[I_a]$. We optimize with GRPO (Shao et al., 2024), which samples n responses per prompt and broadcasts the group-standardized reward to every token.

Token-level entropy control. Hao et al. (2026) leave the advantage untouched and multiply each token's clipped surrogate by a weight $\alpha_t \in [\lambda_{\min}, 1]$ derived from the first-order change in the conditional entropy at a *fixed* state after one step,

$$\Omega(s) \propto -\mathbb{E}_a\Big[\frac{\mathbb{I}_{\text{clip}} A}{\pi_{\text{old}}}\,\pi_\theta(a)\big(1-\pi_\theta(a)\big) \cdot \big(\log \pi_\theta(a) + H\big)\Big], \tag{1}$$

evaluated at the sampled token and mapped into the band by $\alpha_u = \exp(-\alpha\,|\Omega_u| / \max_v |\Omega_v|)$, $\lambda_{\min} = e^{-\alpha}$, so the weight can only attenuate.

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning of reasoning models is well documented and tied to the stall of improvement (Cui et al., 2025); it is countered by an asymmetric clip (Yu et al., 2025), by restricting updates on high-covariance tokens (Cui et al., 2025), or by KL control with reference resets over long training (Liu et al., 2025).

Entropy regularization. Maximum-entropy reinforcement learning, which grew out of the maximum-entropy principle in inverse RL (Ziebart et al., 2008), adds the policy entropy to the objective (Haarnoja et al., 2018). In RLVR a naive

entropy bonus fails to cure the collapse because the large action space and long responses easily trigger a global entropy explosion, which motivates regularizing selectively (Jiang et al., 2025); in continuous control, entropy maximization can mislead policy optimization on tasks that require precise, low-entropy policies (Zhang et al., 2025a). Both bonus and KL act on conditional distributions at visited states, that is, on the local channel.

Token-level intervention. Wang et al. (2025) show that training on a minority of high-entropy "forking" tokens suffices, which motivates per-token treatment but selects tokens by local entropy. Other schemes shape advantages or rewards with token entropy (Cheng et al., 2025; Tan and Pan, 2025; Zhang et al., 2025b). Hao et al. (2026) estimate each token's first-order effect on conditional entropy, the estimator our analysis extends.

Tree-structured rollouts. Sampling a group as a tree of shared prefixes has been used to give intermediate decisions their own reward credit, by estimating the expected outcome below each node (Hou et al., 2025; Yang et al., 2025; Cao et al., 2026), and to choose where to branch: at high-uncertainty positions (Hou et al., 2025; Cao et al., 2026), or where a branch most reduces uncertainty about the policy gradient (Khomich et al., 2026). We use the same structure for a different quantity: siblings supply a baseline for the remaining entropy rather than for the reward, and the schedule is fixed.

Trajectory-level diversity. Token-level entropy can stay high while the diversity of final answers falls (Wu et al., 2025), and explicit outcome-level exploration targets that diversity directly (Song et al., 2025). The visitation channel is a mechanism for the first observation: per-state entropies can be preserved while the trajectory distribution concentrates.

Prior work either controls conditional entropy at visited states or selects tokens by local uncertainty. We instead differentiate the entropy of the trajectory distribution itself, which reveals a term caused by changes in which states the updated policy reaches. The identity this rests on is not new: the trajectory entropy is the entropy term of the maximum-entropy RL objective, and its policy gradient, which credits per-state entropy through a reward-to-go, is standard there (Schulman et al., 2017a; Levine, 2018). What we add is its use to locate token-level entropy control: current estimators measure only the local term, and Section 3.1 shows that this term vanishes where the other is largest. We use Medusa-style heads (Cai et al., 2024) not for decoding but as an entropy forecaster. Appendix A expands this section.

3 Two Channels

The autoregressive factorization and the tower rule give the identity

$$\mathcal{H}(\pi_\theta \mid x) = \mathbb{E}_{y\sim\pi_\theta}\Big[\textstyle\sum_t H\big(\pi_\theta(\cdot \mid s_t)\big)\Big]. \quad (2)$$

Both the integrand and the measure $\mathbb{E}_{y\sim\pi_\theta}$ depend on θ. Writing $f_\theta(y) := \sum_t H(\pi_\theta(\cdot \mid s_t(y)))$, the product rule and the score-function identity give, with no approximation,

$$\nabla_\theta \mathcal{H} = \underbrace{\mathbb{E}\big[\nabla_\theta f_\theta(y)\big]}_{\text{(L) local}} + \underbrace{\mathbb{E}\big[f_\theta(y)\,\nabla_\theta \log \pi_\theta(y \mid x)\big]}_{\text{(V) visitation}}. \quad (3)$$

(L) holds the visited states fixed; Eq. (1) is a first-order estimate of it. (V) holds per-state entropies fixed and asks how the probability of reaching each state changes; Ω reads the distribution at a position and never the probability of arriving there. The two can have opposite signs, so driving (L) to zero does not bound $\nabla_\theta \mathcal{H}$. Appendix B gives every step.

3.1 Why the local channel cannot be repaired

One might expect the channels to coincide, since branch points are uncertain positions; in warm-up rollouts branch diversity does correlate with local entropy ($\rho = 0.65$). The inference fails in three steps.

(i) Ω reads a deviation, not a level. Collecting constants, $\Omega \propto c\,\pi_a(1-\pi_a)\,(H - I_a)$, and $H = \mathbb{E}[I_a]$, so the last factor has mean zero: $|\Omega|$ is set by the *spread* of surprisal, not the height of the entropy. *(ii) At a uniform branch point* $\Omega \equiv 0$, since $I_a = \log k = H$ for every a. *(iii) That zero is correct.* The logit derivative of the conditional entropy (Hao et al., 2026, Eq. 20), $\partial H/\partial z_a = -\pi_\theta(a)(\log \pi_\theta(a) + H)$, vanishes for every a at a uniform distribution; perturbing one logit by δ gives $H''(0) = -(k-1)/k^2$, so the local loss is entirely second order. Uniform branch points are locally safe, and an estimator that allocates them no protection is correct for its own objective.

The visitation term's prefactor, derived in Appendix B.5, is $1 - \pi_a$, which tends to 1 as the

distribution	H	$P(\text{split})$	$\mathbb{E}\lvert\Omega\rvert/\lvert A\rvert$	$\mathbb{E}[1-\pi_a]$
q=.99	0.10	0.08	0.092	0.020
q=.9, tail	1.25	0.57	**1.129**	0.190
q=.6, m=3	1.11	0.98	0.457	0.587
unif. k=10	2.30	**1.00**	**0.000**	**0.900**
unif. k=100	4.61	**1.00**	**0.000**	**0.990**

Table 1: The two channels are oppositely ordered around flat branch points. Top token has mass q, the rest spread uniformly over m tokens; $P(\text{split})$ is the probability that n=8 samples do not all agree.

distribution flattens — where a group of samples is also certain to diverge. Table 1 evaluates both on a family of distributions from confident to flat: $|\Omega|$ is largest where the model is confident but a tail token was sampled, and is zero where divergence is certain and the visitation prefactor is maximal. The local signal vanishes precisely in the regime where the visitation prefactor is largest, so no refinement of the local term recovers what it omits.

4 Estimating the Visitation Channel

A reward-to-go. Expanding (V) into a double sum over the position t of an entropy and the position u of a score, every term with $u \geq t$ is exactly zero: $H(\pi_\theta(\cdot \mid s_t))$ is measurable given $y_{<u}$ and the expected score at a position vanishes. Only $u < t$ survives, and

$$(\mathrm{V}) = \mathbb{E}\Big[\textstyle\sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)\, H_{\text{togo}}(s_u \oplus y_u)\Big], \tag{4}$$

with $H_{\text{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_\theta(\cdot \mid s_t))$. H_{togo} was not posited: it is the reward-to-go of this REINFORCE estimator. Any function of s_u alone is an admissible baseline, giving

$$A_H(s_u, y_u) := H_{\text{togo}}(s_u \oplus y_u) - \bar{H}_{\text{togo}}(s_u), \tag{5}$$

and up to here no approximation has been made. Our baseline is the mean of H_{togo} over rollouts in the same group whose prefixes $y_{<u}$ agree exactly; with no sibling it equals the rollout's own value and $A_H = 0$.

Siblings, and a ceiling. The forecast below is a deterministic function of the causal prefix, so

$$A_H[i,u] \neq 0 \iff \begin{array}{l}\text{siblings share } y_{<u} \\ \text{but diverge at } y_u.\end{array} \tag{6}$$

Independent samples from a sharpened policy diverge at their first token and never share a prefix again, so the visitation channel is identically absent.

We therefore schedule the sharing: a tree sampler cuts a root at fixed depths with branching factor 2, yielding $n = 8$ leaves. The depths are fixed rather than chosen, and we make no claim for the schedule as a sampler. Since n sequences form a trie with at most $n-1$ branching nodes, the fraction of positions with $A_H \neq 0$ is at most $2(n-1)/T$ under *any* sampler — about one percent at $T \approx 10^3$.

The correction cannot move the mean. On each sibling set $\sum_j A_H(s_u, y_u^{(j)}) = 0$. A correction built on A_H therefore cannot change the average weight at branch tokens, only spread siblings apart; this is forced by unbiasedness, not a defect. "Mean entropy at branch points rose" is *not* a prediction of this correction, and an average over all response tokens should barely register it: aggregate token entropy is not a diagnostic for whether the correction acts. The prediction it does make is that the spread *across siblings* widens. A hypothesis that raises the mean is different and weaker, and we run it as a control.

From A_H to a token weight. H_{togo} involves states the update has not visited, so we forecast it from the hidden state with $\kappa = 2$ parallel multi-token-prediction heads, discounted by γ_H and calibrated per head:

$$\hat{H}_{\text{togo}} = \max\Big(0, \textstyle\sum_{k=1}^{\kappa} \gamma_H^k \big(a_k H_k(h_u) + b_k\big)\Big). \tag{7}$$

The head substitution has a known sign: a head models the marginal of a token k steps ahead, whose entropy exceeds the conditional one by a mutual information that grows with k, so the forecast always over-estimates and the affine calibration absorbs it (fitted $a_1 = 1.03$, $a_{k\geq 2} \leq 0.19$). Under the base method's Assumption A, one step moves the sampled token's log-probability by $\eta w_u(1-\pi_u)$, where w_u is the clipped advantage coefficient, giving the visitation prediction $\text{visit}_u = \eta w_u(1-\pi_u)\,\text{clip}(A_H, \pm c)$. When mass moves toward a branch with $A_H > 0$ trajectory entropy rises; when it moves toward one with $A_H < 0$, into a dead end, the abandoned siblings' diversity is destroyed — the collapse path (L) cannot see. Because Ω is heavy-tailed, we add the visitation term *after* the base map, as a bounded adjustment on the support $\mathcal{S} = \{u : \text{visit}_u \neq 0\}$:

$$\alpha_u^{\mathrm{f}} = \text{clamp}\big(\alpha_u + \lambda\, b\, \tanh(\text{visit}_u/\rho),\ \lambda_{\min}, 1\big), \tag{8}$$

with b the band width and ρ the RMS of visit over $\mathcal{S}$. The $\tanh$ preserves the sign and caps one token's movement at λb; at $\lambda = 0$ the pipeline is bitwise the base method. Table 2 lists every approximation and Appendix B.4 derives each. The strongest are A2, whose error has a known sign and monotonicity, and B2, which is false for a network with shared parameters and is inherited from the base method.

5 Experimental Setup

Training. All runs fine-tune Qwen2.5-Math-1.5B (Yang et al., 2024) on DAPO-Math-17k (Yu et al., 2025) for 110 steps of 512 prompts with GRPO, $n = 8$, constant learning rate $1\mathrm{e}{-6}$, responses up to 3072 tokens, no entropy bonus or KL loss, and an asymmetric clip ($\varepsilon_{\text{low}} = 0.2$, $\varepsilon_{\text{high}} = 0.28$). The base method's map uses $\lambda_{\min} = 0.7$, its paper's value (Hao et al., 2026, App. F.1). The correction uses $\kappa = 2$, $\gamma_H = 0.7$, $\lambda = 0.25$, $c = 1$, and a tree cut at depths $\{64, 192, 384\}$. Every arm shares the data seed per run and the initial checkpoint. Appendix D lists the rest.

Five arms. We call everything except the content of A_H — the tree sampler, the support $\mathcal{S}$ and the token-weight damping — the *damping mechanism*. Comparing the base method with the arm that applies the full correction changes the mechanism and the signal at once, so we add two arms that keep the whole mechanism and remove only the branch-specific A_H signal (Table 3). UNIFORM applies the same attenuation to every position in the support: it removes the A_H signal while preserving its support, which implements the "raise the mean" hypothesis. PERMUTED shuffles A_H within each sibling set: it preserves the distribution of A_H magnitudes and the zero sum but destroys the branch–score assignment. SIGNED applies the correction as derived. It is the instrument with which the analysis's predictions are tested; we do not propose it as a training method.

Statistics. Every run validates on AIME24 (30 problems, 32 samples) every 10 steps; one evaluation has an across-problem standard error of .0282. The primary statistic is the mean over the converged window, steps 40–110. The seed is the unit of analysis: each run is collapsed to its window mean, arms are paired within a seed, and the test is a paired t with $\mathrm{df} = n - 1$, over the 3 seeds on which all five arms finished $(1, 3, 4)$. We report acc@32, maj@32 and their difference (uplift). The logged entropy is a token mean, i.e. $\mathcal{H}$ divided by mean length. The analysis plan was registered before the runs (Appendix I), and every figure is recomputed from the training logs by one script.

6 Results

The results are ordered by what the analysis predicts. The first is the dissociation the zero sum implies (Section 6.1); the per-arm numbers, accuracy included, come after it as the material the contrasts are built from.

6.1 Aggregate entropy tracks the damping mechanism, not the A_H signal

Question. Does aggregate entropy respond to the A_H signal itself? The construction says it should not: A_H is non-zero on at most $2(n-1)/T$ of positions and redistributes weight among siblings without adding any (Section 4). Both statements were in the manuscript before the campaign was launched.

Result. Across all three paired seeds, changing the damping mechanism produces substantially larger shifts in aggregate entropy than changing the A_H signal while holding the mechanism fixed (Table 4). Turning the damping on shifts it by $-.0419$, and adding the tree with uniform damping by $+.0184$; removing the A_H signal shifts it by $-.0077$, and destroying the branch–score assignment by $+.0044$. The smaller mechanism effect is more than twice the larger signal effect. Accuracy runs the other way: the signal contrasts give $+.0069$ and $+.0087$, the mechanism contrasts $-.0065$ and $+.0051$. The split survives length correction: on the trajectory entropy $\mathcal{H}$ itself the mechanism contrasts are -42.5 and $+16.6$ nats per response and the signal contrasts -6.4 and $+1.8$ (Appendix Table 9).

Interpretation. Aggregate entropy primarily reflects the damping mechanism rather than the branch-specific A_H signal, which is the dissociation the zero sum predicts and the sense in which this is not a method for suppressing entropy.

Caveat. Three paired seeds establish the qualitative pattern more strongly than its effect sizes; a small difference that reproduces across seeds can still reach a large t, as the PERMUTED row does. The positive prediction — that the spread across siblings widens — is not measured; a registered

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (2)–(5)	nothing — identities	—	—
A1	horizon κ, discount γ_H	the tail $\sum_{k>\kappa}$, and a fraction of the retained terms	under-estimate	the common component cancels in the sibling difference
A2	realized future $\to$ MTP marginal	mutual information with intervening tokens, Eq. (26)	**over-estimate, monotone in** k	A3
A3	per-head affine calibration	non-affine residual of A2	least-squares residual	oracle-forecast control arm
A4	clip A_H to $\pm c$	magnitude of extreme branch scores	bounded by c	saturation rate is logged
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of weight sharing	—	*inherited from the base method*
B3	rollouts come from π_{old}	mismatch with the π_θ expectation	—	importance ratio restores r
B4	self included in sibling set	$O(1/m)$ relative to leave-one-out	—	same property as the GRPO advantage

Table 2: Every approximation between the exact decomposition and the implemented weight. A1–A4 are introduced by this work; B1–B4 are shared with the token-level entropy-control method we extend.

arm	λ	rollout	signal in the weight
GRPO	—	plain	none
STEER	0	plain	local Ω only
uniform	.25	tree	support only
permuted	.25	tree	A_H shuffled
signed	.25	tree	A_H as derived

Table 3: The five arms. UNIFORM and PERMUTED keep the damping mechanism and remove the branch-specific A_H signal in two different ways.

contrast	n	Δent.	t	Δacc	t
the damping mechanism changes, the A_H signal does not					
STEER − GRPO	3	−.0419	−5.87	−.0065	−1.24
UNIF − STEER	3	+.0184	+13.09	+.0051	+1.37
the A_H signal changes, the damping mechanism does not					
SIGNED − UNIF	3	−.0077	−1.42	+.0069	+1.16
SIGNED − PERM	3	+.0044	+2.15	+.0087	+3.11
both change					
SIGNED − STEER	3	+.0108	+1.76	+.0120	+5.24
SIGNED − GRPO	3	−.0312	−14.01	+.0055	+0.81

Table 4: Converged entropy and accuracy, paired within a seed ($\mathrm{df} = n - 1$) and grouped by what the contrast changes. Majority-vote contrasts are in Appendix Table 10.

condition on the ordering of arm means was retired rather than scored, for reasons given in Appendix E.3.

6.2 Accuracy and entropy by arm

Table 5 averages each arm over the converged window, and Table 6 gives one seed's full trajectory. The three arms without the A_H signal — STEER, UNIFORM, PERMUTED — sit below GRPO on accuracy and SIGNED above it, by less than GRPO's

arm	n	acc@32	maj@32	uplift	entropy
GRPO	3	.1436 ± .0064	.2145 ± .0100	.0710	.1575
STEER	3	.1371 ± .0021	.2090 ± .0054	.0719	.1155
UNIFORM	3	.1422 ± .0057	.2114 ± .0093	.0692	.1340
PERMUTED	3	.1405 ± .0030	.2129 ± .0023	.0724	.1219
SIGNED	3	.1491 ± .0005	.2301 ± .0042	.0810	.1263

Table 5: Means over steps 40–110 on seeds $1, 3, 4$, $\pm$ the standard error across seeds. Because the seed set is shared, a difference of two rows is the paired contrast of Table 4. Entropy is the token-mean quantity.

standard error across seeds. On entropy GRPO holds the most and the four damped arms sit close together. The two columns are neither aligned nor reversed; read as the contrasts of Section 6.1, they separate.

6.3 Does the branch-specific A_H signal matter for accuracy?

The pre-registered test. The registration made SIGNED − UNIFORM the accuracy test: *if it straddles zero across seeds, the A_H signal does not matter*. It straddles: over 3 paired seeds it runs from $-.0020$ to $+.0182$, with 1 negative, and we do not rescue it with the mean or with majority vote. The pre-registered accuracy hypothesis is therefore not supported, and we make no claim that A_H improves accuracy. The other three registered conditions did not trigger: SIGNED − PERMUTED, which keeps the magnitudes and destroys only the branch–score assignment, does not straddle ($+.0033$ to $+.0125$); the arms are not ordered alike by entropy and accuracy; and the mean token weight stays at .9990.

step	GRPO	STEER	UNIF	PERM	SIGNED
0	.039/.035	.043/.040	.039/.035	.039/.035	.039/.035
10	.075/.127	.064/.109	.070/.124	.062/.126	.078/.148
20	.120/.176	.114/.195	.102/.171	.087/.155	.103/.181
30	.108/.195	.135/.215	.121/.223	.099/.178	.113/.211
40	.144/.236	.115/.187	.119/.201	.116/.197	.141/.230
50	.120/.204	.125/.196	.121/.216	.153/.224	.145/.258
60	.126/.178	.139/.202	.125/.203	.135/.190	.145/.237
70	.130/.200	.128/.200	.129/.197	.132/.227	.147/.257
80	.135/.182	.135/.202	.140/.204	.134/.197	.144/.206
90	.136/.191	.144/.212	.142/.178	.130/.210	**.166**/.239
100	.132/.171	.146/.207	.134/.192	.144/.212	.157/.222
110	.157/.208	.132/.181	.140/.171	.152/.218	.151/.230

Table 6: AIME24 acc@32/maj@32 at every validation point, seed 1. Rows below the rule are the converged window used for all subsequent statistics. One seed is one trajectory: the across-seed statistics are in Tables 5 and 10.

The remaining gaps. Against STEER the gap is $+.0120$ ($p = 0.035$), but STEER lands below GRPO here although its paper places it above, a discrepancy we could not attribute (Appendix E.10). Against GRPO the gap of $+.0055$ is a fraction of GRPO's spread across seeds, .0210, and is $+.0011$ on the seeds where equal steps are equal compute (Appendix E.11). On six benchmarks the registered secondary check counts SIGNED above UNIFORM on 3 of six, which its thresholds call *equivocal* (Appendix E.9).

6.4 What the sampler and the forecaster contribute

Single-seed follow-ups (Table 7) locate the rest of the effect. *The sampler:* the tree with the forecast switched off ($\lambda = 0$) reaches .1439 against STEER's .1371 and SIGNED's .1491, and an uplift of .0870 against SIGNED's .0810, so most of the majority-vote structure belongs to the sampler; we report this and leave it to work on structured sampling, since the tree is here only to give A_H siblings. *The forecaster:* on a frozen policy, the same sibling operator applied to the policy's own next-token entropy reproduces A_H at $R^2 = 0.840$ (Appendix E.7); an ORACLE arm reading H_{togo} from realized entropy is 19.3% cheaper per step and $+.0112$ behind at one seed, so we keep the heads without claiming they are necessary. *The support:* the tree arms carry a non-zero A_H on .0125 of positions against a ceiling of .0143. On the four-GPU box a SIGNED step costs $+10\%$ more than STEER's (Appendix E.12).

arm	what changes	acc@32	maj@32
the damping coefficient			
$\lambda = 0.1$	weaker	.1365	.2360
$\lambda = 0.25$	as used	.1491	.2301
$\lambda = 0.5$	stronger	.1444	.2265
the sampler, with the forecast switched off			
$\lambda = 0$, tree	tree only	.1439	.2309
STEER	neither	.1371	.2090
the forecaster, replaced by what the trainer already has			
ORACLE	no heads	.1369	.2234
relaxed ratio clipping (ϵ_{hi}=5, ϵ_{lo}=0.99)			
SIGNED	—	.1516	.2304

Table 7: Follow-up ablations, converged window, one seed per row and no error bar. The seed is whichever one the arm was run at, and our analysis emits it with the row; the ORACLE arm is seed 4, run there so that it shares a seed with the SIGNED and UNIFORM runs the six-benchmark evaluation pairs. The $\lambda = 0.25$ line is the reference and is Table 5's across-seed mean rather than one run. Every row but ORACLE ran on the 4-GPU box while the seed-1 SIGNED reference ran on the smaller one, so those rows carry the machine effect of Appendix E.8 ($+.0129$, in the direction that flatters them) on top of whatever the ablation does. Our analysis emits a same-seed contrast for ORACLE alone, because it is the one row whose box matches its reference's. Each block isolates one element of the design. The relaxed-clipping block has no STEER control: that run stopped at training step 62, before the converged window closes, and a mean over the part of the window it reached is not the quantity the other rows report. Arms that were planned and did not produce a finished run are listed in Appendix I.

6.5 Calibration against published numbers

To check that our pipeline lands in the range the base method's paper reports, we also ran the SIGNED configuration under that paper's protocol (Hao et al., 2026, App. E): two independent 200-step runs (data seeds 1, 2), AIME24 validation every 10 steps, and the AIME24-best checkpoint of each run evaluated on six benchmarks and averaged. The tree uses two roots cut at $\{64, 192\}$, chosen after the five-arm campaign on training prompts to keep the siblings while making fewer rewards in a group tie; the choice is exploratory. Table 8 sets the result beside the base method's published 1.5B table.

The SIGNED arm's two-run mean is 36.7, against 38.2 for the base method and 35.8 for GRPO as reported: lower than the former on 5 of six benchmarks (not on AIME24, the selection column) and higher than the latter on 5. The quoted rows were not re-run and used different hardware and code, so the comparison places the corrected arm be-

	AIME24†	AIME25	AMC23	MATH500	Minerva	Olympiad	mean
reported by Hao et al. (2026), not re-run							
base model	4.1	2.1	24.7	29.0	9.2	20.5	14.9
GRPO	16.2	7.6	56.0	74.4	26.1	34.6	35.8
STEER	17.4	9.7	61.6	75.5	28.2	36.6	38.2
ours, mean of the two runs							
SIGNED, selected	17.4	8.3	57.6	73.8	27.0	35.7	36.7
SIGNED, step 200	16.9	9.5	59.6	73.9	28.9	36.0	37.5
ours, each run's selected checkpoint							
seed 1	18.0	9.2	57.1	74.0	24.6	34.8	36.3
seed 2	16.9	7.5	58.2	73.6	29.4	36.6	37.0

Table 8: Six benchmarks under the base method's protocol, accuracy in percent; the mean column is unweighted over the six. The upper block is quoted from the base method's paper and was not re-run. In each of our runs the selected checkpoint is the argmax of its AIME24 validation accuracy. †: in our rows AIME24 is the training-time validation value at the selected step — the value the selection itself used — and is biased upward by that selection; the other five columns re-evaluate the selected checkpoint. The base method's paper selects the same way and evaluates the selected checkpoint, but does not say whether its AIME24 column is the selection value or a re-measurement; if it is a re-measurement, this column favours our rows. Seed 2's selected checkpoint was evaluated on one GPU (tensor parallel size 1), every other checkpoint of ours on four, because the four-GPU attempt stopped before producing a metric; the sampling settings are the same.

tween the two reported rows without ranking it (Appendix F).

7 Conclusion

Current token-level entropy control in RLVR accounts for how visited states sharpen, but not for how policy updates change which states are visited. The gradient of trajectory entropy has both terms; the local estimator vanishes, for a correct reason, exactly where the visitation term is largest, so the omission cannot be repaired by a better local estimator. The missing term is a REINFORCE estimator whose reward-to-go is the remaining conditional entropy, and a fixed tree supplies the sibling baseline that makes it estimable. The resulting correction is mean-preserving, so it can change trajectory diversity without a corresponding change in aggregate token entropy, and the experiment observes that dissociation: aggregate entropy tracks the damping mechanism, not the A_H signal. We do not establish an accuracy benefit, and this paper should be read as an account of where token-level entropy control acts and of what an aggregate entropy curve cannot see, not as a recommendation to adopt the correction for accuracy. Whether a cheaper estimator of H_{togo} suffices, and whether structured sampling helps RLVR in its own right, remain open.

Limitations

Empirical scope. Every result is measured on one 1.5B mathematics-specialized model, on 3 paired seeds $(1, 3, 4$; the registration planned five, Appendix I), and validated on AIME24's 30 problems, where one evaluation carries an across-problem standard error of .0282. The seed count bounds effect sizes rather than the structural result, which is a comparison between groups of contrasts; but the accuracy gaps are inside the spread across seeds — GRPO's converged accuracy moves by .0210 across its seeds — and the pre-registered accuracy test of the A_H signal is not supported (Section 6.3). The six-benchmark check covers one seed per arm. The base method lands below GRPO in our runs, where its paper reports it above; we ran it at its paper's $\lambda_{\min} = 0.7$ rather than its launcher's 0.8, on 110 rather than 200 steps, and cannot separate these causes (Appendix E.10). This reaches SIGNED − STEER and STEER − GRPO, not the contrasts against UNIFORM and PERMUTED, which hold every setting fixed. The comparison with published numbers (Section 6.5) is uncontrolled: the quoted rows were not re-run, our AIME24 column is the selection value and biased upward, and the sampler in those runs was chosen after the campaign.

Estimator assumptions and efficiency. The calibration suppresses heads beyond the first, so $\hat{H}_{\text{togo}}$ is close to a one-step forecast, and on a frozen policy the policy's own next-token entropy reproduces A_H at $R^2 = 0.840$; the heads carry the step-time overhead of Appendix E.12. The in-loop ORACLE comparison favours the heads by $+.0112$ at one seed, which orders the two without sizing the difference; whether the heads earn their cost is open. Approximation B2, that an update does not change the conditional distribution at other positions, is false for a network with shared parameters; it is inherited from the base method, but our construction adds a second estimator that rests on it. The support is small by construction — at most $2(n-1)/T$, about one position in eighty here.

Open structural predictions. The prefactor half of Section 3.1 is analytic, but we have not measured the local and visitation terms together at real sibling-divergence positions in live rollouts, so Ta-

ble 1 is a prediction rather than an observation. The positive prediction of the zero sum — that the spread across siblings widens — is not measured: no run logs the realized to-go entropy of siblings at their branch points. The dissociation we measure is its negative half. Whether the effect scales with the number of samples or with an adaptively chosen branching schedule (Cao et al., 2026; Khomich et al., 2026) is untested, and whether structured rollout sampling improves RLVR in its own right is a question this paper raises and does not answer.

Ethical Considerations

This work studies an optimization-time regularizer for reinforcement learning on mathematics problems with automatically verifiable answers. It introduces no new data collection, no human subjects, and no new model release. The intervention acts on which of several equally-scored continuations receives a slightly smaller gradient weight; it does not alter the reward, and so does not change what behavior is being optimized for. The training data is a public competition-mathematics set and the evaluation set is a public examination.

The main foreseeable risk is indirect. Methods that preserve diversity during RL fine-tuning are not specific to mathematics, and the same mechanism would apply to a reward model encoding any objective, including a harmful one; we note that our correction inherits, and does not mitigate, whatever the reward function encodes. The computational cost of the correction (a forward pass through a set of small prediction heads at every optimization step) is a real environmental cost, which we quantify in Appendix E.12 so that it can be weighed against the benefit.

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A Extended Related Work

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning is well documented (Cui et al., 2025), and a range of interventions add an explicit entropy term to the objective, in the tradition of maximum-entropy reinforcement learning (Haarnoja et al., 2018), itself rooted in the maximum-entropy principle of inverse RL (Ziebart et al., 2008), or constrain the update so that high-entropy positions are preserved. Recent analyses report that naive entropy regularization fails to cure the collapse in large reasoning models, because the vast action space and long trajectories easily trigger a global entropy explosion, and they regularize selectively instead (Jiang et al., 2025), which is consistent with our reading: the regularizer acts on the local channel, and the local channel is not where trajectory diversity is being lost.

Assumption. Both Hao et al. (2026) and this work rely on one structural assumption, which we restate so that our approximation ledger can refer to it.

> **Assumption 1** (parameter-independent softmax; Hao et al., 2026). For any state s, each action a carries an independent logit $z_{s,a}(\theta)$, so that a gradient step on the sampled token does not substantially move the logits of the other tokens at that state.

It is false for a network with shared parameters. We inherit it rather than introduce it, but our construction adds a second estimator that rests on it.

Token-level entropy control. The method we build on re-weights the clipped surrogate loss by a per-token first-order estimate of the entropy change that token's update induces (Hao et al., 2026). That work observes that earlier interventions are "indirect" and that their effect is therefore limited and can fail. Our decomposition supplies a mechanism for that observation: the estimate is a first-order estimate of the local channel only, and it is exactly zero at the states where the visitation channel is largest.

Which tokens carry the gradient. Wang et al. (2025) show that restricting policy-gradient updates to a small minority of high-entropy "forking" tokens matches full-gradient training. That result establishes that the identity of the tokens, not the aggregate entropy, is what matters, and it motivates a per-token treatment; it does not, however, distinguish the two channels, since forking tokens are identified by their local entropy. Our analysis predicts that a local criterion will select a systematically different set from the one the visitation channel would select.

Tree-structured rollouts. TreeRL (Hou et al., 2025) trains with on-policy tree search, branching from high-uncertainty intermediate steps, and derives step-level credit from the outcomes below each node; TreeRPO (Yang et al., 2025) forms step-level groups from tree sampling to estimate the expected reward of intermediate steps. TreeAdv (Cao et al., 2026) branches where the token entropy exceeds a threshold and assigns each shared segment the mean of the group-normalized advantages of the leaves below it. EPIG-Tree (Khomich et al., 2026) places branches where an additional branch most reduces uncertainty about the policy gradient. A_H has the same shape as these node-level estimates — an average over the continuations below a node, compared with a reference — but its reward-to-go is the remaining conditional entropy, its reference is the sibling mean, so it sums to zero over each sibling set, and it enters through the token weight rather than through the advantage. Whether an adaptive schedule of the kind these methods use changes what the visitation term does is left open.

Local stochasticity without global diversity. Wu et al. (2025) report that token-level entropy can remain high while the diversity of final answers falls. This is precisely the signature our decomposition predicts when the visitation channel is unmanaged: per-state entropies are preserved — the local channel is doing its job — while the trajectory distribution concentrates.

Maximum-entropy policy gradients. Treating per-state entropy as a reward is the maximum-entropy formulation of RL (Ziebart et al., 2008; Levine, 2018), and its policy gradient already contains both terms of Eq. (3): the entropy at a state is credited to earlier actions through a reward-to-go, which is our visitation channel, alongside the direct gradient of the entropy at the visited state, which is our local channel (Schulman et al., 2017a). We claim no novelty for the identity. The contribution is to apply it to the token-level estimators used in RLVR, which keep only the second term, and to show (Section 3.1) that the term they keep is zero exactly where the term they drop is largest.

Multi-token prediction. We use Medusa-style parallel prediction heads (Cai et al., 2024) not for their intended purpose (speculative decoding) but as an entropy forecaster: we need the entropy of the distribution over a token k steps ahead, not a good sample of that token. Appendix B.4 shows that this choice of estimator has a signed, monotone error which we characterize and correct.

B Full Derivations

This appendix gives the derivations that Sections 2–4 state in brief.

B.1 Preliminaries

Notation. A prompt is x, a response is $y = y_{1:T}$, and the state before emitting token t is $s_t = (x, y_{<t})$. We write $H(\pi_\theta(\cdot \mid s_t))$ for the entropy of the conditional distribution at a single position (lowercase, per-token) and $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution over whole responses (uppercase, per-trajectory). The surprisal of a token a is $I_a := -\log \pi_\theta(a \mid s_t)$, so that $H = \mathbb{E}_{a\sim\pi_\theta}[I_a]$.

Group-relative policy optimization. We optimize with GRPO. For each prompt, n responses are sampled and scored, and each response receives the group-standardized advantage $A_i = (r_i - \mathrm{mean}_j r_j)/\mathrm{std}_j r_j$, broadcast to every token of that response. Replacing a learned value function with the group mean is what makes GRPO cheaper than PPO; it also means that a group whose members all receive the same reward contributes no gradient.

Token-level entropy control. Hao et al. (2026) leave the advantage untouched and instead multiply each token's contribution to the clipped surrogate by a weight $\alpha_t \in [\lambda_{\min}, 1]$ derived from a first-order estimate of the entropy change that token's update induces. Their Theorem 1 gives that estimate in closed form: under a parameter-independent softmax (their Assumption 1; our A), the change in the conditional entropy at a *fixed* state s after one GRPO step is, to first order,

$$\Omega(s) = -\frac{\eta}{L}\,\mathbb{E}_a\Big[\frac{\mathbb{I}_{\mathrm{clip}}(s,a)\,A(s,a)}{\pi_{\mathrm{old}}(a \mid s)} \cdot \pi_\theta(a \mid s)\big(1-\pi_\theta(a \mid s)\big) \cdot \big(\log \pi_\theta(a \mid s) + H\big)\Big], \tag{9}$$

where η is the learning rate, L the total decoded length in the update, and $\mathbb{I}_{\mathrm{clip}}$ the PPO clipping indicator. The implementation evaluates the bracket at the sampled token rather than in expectation. The policy loss becomes $\sum_u \alpha_u \cdot \text{PPO-clip}(A_u, r_u)$, with

$$\alpha_u \;=\; \exp\Big(-\alpha\,\frac{|\Omega_u|}{\max_v |\Omega_v|}\Big), \qquad \lambda_{\min} = e^{-\alpha}, \tag{10}$$

a batch-normalized exponential map into $[\lambda_{\min}, 1]$, so the weight can only attenuate. The factor $1/\pi_{\mathrm{old}}$ in Eq. (1) will matter later: it makes Ω heavy-tailed, which is why the two channels must be normalized before they are combined (Appendix B.6).

B.2 Two Channels

B.2.1 Step 1: the chain rule

The object we would like to protect is the entropy of the distribution the policy induces over *whole responses*,

$$\mathcal{H}(\pi_\theta \mid x) := -\sum_y \pi_\theta(y \mid x)\,\log \pi_\theta(y \mid x), \tag{11}$$

a sum over an exponentially large set. The autoregressive factorization $\pi_\theta(y \mid x) = \prod_t \pi_\theta(y_t \mid s_t)$ turns the logarithm into a sum and lets us exchange the order of summation:

$$\begin{aligned}\mathcal{H} &= -\sum_y \pi_\theta(y \mid x) \sum_{t=1}^{T} \log \pi_\theta(y_t \mid s_t) \\ &= \sum_{t=1}^{T} \mathbb{E}_{y\sim\pi_\theta}\big[-\log \pi_\theta(y_t \mid s_t)\big].\end{aligned} \tag{12}$$

Each summand is still an expectation over whole responses, but the integrand depends on y only through $y_{\le t}$. Conditioning on the prefix and applying the tower rule replaces the inner expectation with a per-position entropy,

$$\mathbb{E}_y\big[-\log \pi_\theta(y_t \mid s_t)\big] = \mathbb{E}_{y_{<t}}\Big[\underbrace{\mathbb{E}_{y_t\sim\pi_\theta(\cdot|s_t)}\big[-\log \pi_\theta(y_t \mid s_t)\big]}_{=\,H(\pi_\theta(\cdot|s_t))}\Big],$$

because the inner expectation is, by definition, the entropy of the conditional distribution at s_t. Collecting the terms back under a single expectation gives

$$\mathcal{H}(\pi_\theta \mid x) = \mathbb{E}_{y\sim\pi_\theta}\Big[\sum_{t=1}^{T} H\big(\pi_\theta(\cdot \mid s_t)\big)\Big]. \tag{13}$$

Equation (2) is an identity, not an approximation. Its left-hand side is what we want to protect; its right-hand side is what instrumentation can measure, since $H(\pi_\theta(\cdot \mid s_t))$ is read directly off the logits at each position.

The entire argument of this paper follows from one feature of Eq. (2) that is easy to pass over: *the measure $\mathbb{E}_{y\sim\pi_\theta}$ depends on θ as well*. The right-hand side is not a sum of θ-dependent terms over a fixed set of positions — it is a θ-dependent average of them over a θ-dependent set of trajectories.

B.2.2 Step 2: the product rule

Write the per-trajectory total

$$f_\theta(y) := \sum_{t=1}^{T} H\big(\pi_\theta(\cdot \mid s_t(y))\big), \tag{14}$$

so that Eq. (2) reads $\mathcal{H} = \sum_y \pi_\theta(y \mid x)\, f_\theta(y)$. Both factors of every summand carry θ, so differentiating requires the product rule:

$$\nabla_\theta \mathcal{H} = \sum_y \big[\nabla_\theta \pi_\theta(y \mid x)\big] f_\theta(y) + \sum_y \pi_\theta(y \mid x)\,\nabla_\theta f_\theta(y). \tag{15}$$

The second sum is already an expectation. The first is not, because $\nabla_\theta \pi_\theta$ is not a probability distribution; the score-function identity $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$ restores it to one. Substituting and naming the two terms,

$$\nabla_\theta \mathcal{H} = \underbrace{\mathbb{E}\big[\nabla_\theta f_\theta(y)\big]}_{\text{(L) local}} + \underbrace{\mathbb{E}\big[f_\theta(y)\, \nabla_\theta \log \pi_\theta(y \mid x)\big]}_{\text{(V) visitation}}. \tag{16}$$

Equations (11)–(3) contain no approximation of any kind. The split into two channels is forced by the product rule, not chosen.

B.2.3 Reading the two terms

(L), the local channel, holds the visited states fixed and asks how the conditional distribution at each one changes. Equation (1) is a first-order estimate of exactly this.

(V), the visitation channel, holds the per-state entropies fixed and asks how the probability of reaching each state changes. An update that concentrates mass on one continuation makes the subtree below it more likely and the sub-trees below its siblings less likely. Nothing in Eq. (1) can see this: Ω reads the conditional distribution at a position and never reads the probability of arriving there.

The two terms can have opposite signs. A method that drives (L) to zero has not thereby bounded $\nabla_\theta \mathcal{H}$.

B.2.4 Why the local channel cannot be repaired

A natural objection is that the two channels might coincide in practice: branch points are positions where the policy is uncertain, so they are high-entropy positions, so Ω should be large there and the local term should protect them automatically. The premise is correct — in warm-up rollouts on our model, branch diversity correlates with local entropy at Spearman $\rho = 0.65$, and with the entropy-to-go at $\rho = 0.82$. The inference from it fails, in three steps.

Step 1: Ω reads a deviation, not a level. Collecting the constants of Eq. (1) into $c := \eta\, \mathbb{I}_{\text{clip}} A / (L\, \pi_{\text{old}})$ and rewriting in terms of surprisal gives, for the contribution of a single action a,

$$\Omega \;\propto\; c\, \pi_a (1 - \pi_a)\, \big(H - I_a\big), \tag{17}$$

and $H = \mathbb{E}_{a \sim \pi}[I_a]$ by definition. The factor $H - I_a$ is therefore a mean-zero fluctuation: what sets the magnitude of $|\Omega|$ is the *spread* of surprisal, not the *height* of the entropy. Nothing in c depends on the shape of $\pi(\cdot \mid s)$, so the argument below is unaffected by the clipping indicator or the importance weight.

Step 2: at a uniform branch point $\Omega \equiv 0$. If the distribution is uniform over k candidates then $I_a = \log k = H$ for every a, so Ω vanishes identically. The positions that are most branch-like — several continuations at comparable probability — are exactly where the local term contributes nothing.

Step 3: and that zero is the correct answer. This is the step that closes the argument, and it follows from Hao et al.'s own derivation. Differentiating the conditional entropy with respect to a single logit gives (Hao et al., 2026, Eq. 20)

$$\frac{\partial H(\pi_\theta \mid s)}{\partial z_{s,a}} = -\,\pi_\theta(a \mid s)\big(\log \pi_\theta(a \mid s) + H(\pi_\theta \mid s)\big), \tag{18}$$

and at a uniform distribution $\log \pi_\theta(a \mid s) = -\log k = -H$ makes the bracket vanish for *every* a. So $\partial H / \partial z_a = 0$ identically: Ω is not wrong, it is right. Perturbing one logit of a uniform distribution by δ and writing $S := e^\delta + k - 1$ gives the closed form

$$H(\delta) = \log S - \frac{\delta e^\delta}{S}, \qquad H''(0) = -\frac{k-1}{k^2}, \tag{19}$$

with $H'(\delta)$ of sign $-\operatorname{sign}(\delta)$: any push in any direction lowers the entropy, but the loss is entirely second order. For a 10-way uniform distribution and $\delta = 0.1$ the actual entropy loss is 4.7×10^{-4}, roughly two orders of magnitude below that of a peaked distribution under the same perturbation. *As far as local entropy is concerned, uniform branch points are genuinely safe*, and a local estimator that allocates them no protection is allocating correctly for its own objective. No refinement of the local term recovers the missing signal, because the missing signal is not local.

The two channels are oppositely ordered. The visitation term's prefactor, derived in Appendix B.5, is $1 - \pi_a$, which approaches 1 as the distribution flattens — and the probability that a group of n samples actually diverges at a position approaches 1 in the same limit. Table 1 evaluates both channels on a family of distributions that interpolates

between confident and flat. $|\Omega|$ is dominated by positions where the model is fairly confident but a tail token was sampled; at those positions siblings diverge only about half the time. Where divergence is certain, the local term is identically zero and the visitation prefactor is maximal.

The statement we can therefore make is stronger than "the local term is incomplete": *the local term is weakest exactly where the term it omits is strongest*, and the omission cannot be fixed within the local channel.

B.3 Estimating the Visitation Channel

B.3.1 Causal elimination gives a reward-to-go

The visitation term of Eq. (3) is not yet in a usable form: the factor $f_\theta(y)$ multiplying the score is a property of the *whole* trajectory, so a naive estimator would credit every token with entropy that appeared before it. Causality removes those terms exactly.

Expanding both factors — f_θ by Eq. (14) and the score by $\nabla_\theta \log \pi_\theta(y \mid x) = \sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)$ — turns (V) into a double sum:

$$(\mathrm{V}) = \sum_t \sum_u \mathbb{E}\big[H(\pi_\theta(\cdot \mid s_t)) \cdot \nabla_\theta \log \pi_\theta(y_u \mid s_u)\big]. \tag{20}$$

Consider a term with $u \geq t$. The entropy $H(\pi_\theta(\cdot \mid s_t))$ depends on y only through $y_{<t}$, and $y_{<t} \subseteq y_{<u}$, so it is measurable with respect to $y_{<u}$ and may be pulled out of a conditional expectation given $y_{<u}$. What remains is the expected score at a single position, which vanishes for any distribution:

$$\begin{aligned}\mathbb{E}\big[\nabla_\theta \log \pi_\theta(y_u \mid s_u) \mid y_{<u}\big] &= \sum_a \pi_\theta(a \mid s_u) \frac{\nabla_\theta \pi_\theta(a \mid s_u)}{\pi_\theta(a \mid s_u)} \\ &= \nabla_\theta \sum_a \pi_\theta(a \mid s_u) \;=\; \nabla_\theta 1 \;=\; 0.\end{aligned} \tag{21}$$

By the tower rule every (t, u) term with $u \geq t$ is therefore *exactly* zero — not small, zero. This is the same identity that licenses a baseline in policy gradients, used here to enforce causality rather than to reduce variance.

Only $u < t$ survives. Swapping the order of summation collects, for each token u, the entropies at all later positions:

$$(\mathrm{V}) = \mathbb{E}\Big[\sum_u \nabla_\theta \log \pi_\theta(y_u \mid s_u)\, H_{\mathrm{togo}}(s_u \oplus y_u)\Big], \tag{22}$$

with $H_{\mathrm{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_\theta(\cdot \mid s_t))$.

We stress that H_{togo} was not posited. Equation (4) is a REINFORCE estimator whose reward is "the conditional entropy remaining after this token", and H_{togo} is that reward-to-go.

B.3.2 The sibling-prefix baseline

Any function $b(s_u)$ of the state alone leaves Eq. (4) unbiased, by the same argument. Subtracting one gives the per-token advantage

$$A_H(s_u, y_u) := H_{\mathrm{togo}}(s_u \oplus y_u) - \bar{H}_{\mathrm{togo}}(s_u), \tag{23}$$

and up to this point *no approximation has been made*.

The baseline we use is the mean of H_{togo} over the rollouts in the same prompt group whose prefixes $y_{<u}$ agree exactly. Those rollouts share the state s_u by definition, so their mean is a function of s_u alone and the baseline is admissible. The condition is agreement on $y_{<u}$ and not on $y_{\leq u}$: the rollouts must diverge *at* u for the difference to be a branch score.

Two properties follow immediately and matter later. First, the implementation includes a rollout in its own sibling set, so b depends weakly on y_u and the estimator is not strictly leave-one-out; the resulting bias is $O(1/m)$ in the number m of surviving siblings, the same property GRPO's group-mean advantage has. Second, when $m = 1$ the baseline equals the rollout's own value and $A_H \equiv 0$, so a position with no siblings is silent rather than noisy.

B.4 Four Approximations

Equation (5) is exact but not computable: H_{togo} sums conditional entropies at states the update has not visited. Three approximations make it computable and a fourth bounds its influence.

B.4.1 A1: horizon truncation and discounting

We truncate the sum to κ steps and discount:

$$H_{\mathrm{togo}}(s_u \oplus y_u) \approx \sum_{k=1}^{\kappa} \gamma_H^k\, H\big(\pi(\cdot \mid s_{u+k})\big). \tag{24}$$

The discount serves two purposes: distant terms carry more forecast error (A2) and therefore deserve less trust, and the sum stays bounded independently of κ.

What this discards — the tail $\sum_{k>\kappa}$ and a fraction of the retained terms — is largely harmless for a reason specific to our use. Equation (5) consumes only the *difference* between siblings. Any

bias common to rollouts sharing s_u cancels in that difference, and truncation bias is common to first order.

B.4.2 A2: forecasting the unrealized future

Even Eq. (24) requires the future states s_{u+k}. At update time they do not exist: rollouts came from π_{old}, each state has exactly one sampled continuation, and re-rolling is prohibitive. We replace the realized future with a parallel forecast from the current hidden state h_u:

$$H\big(\pi(\cdot \mid s_{u+k})\big) \;\longrightarrow\; H\big(p^{(k)}_{\text{MTP}}(\cdot \mid h_u)\big). \quad (25)$$

Head k is a two-layer residual MLP applied to h_u, followed by the policy's own unembedding; the heads own no vocabulary parameters. Their last layer is zero-initialized, so before any training every head reproduces the policy's own next-token distribution and the forecast starts from a sane value. In the RL path the logits are materialized in chunks and reduced to an entropy immediately, so the $[\kappa, B, T, V]$ tensor never exists.

The error of this substitution has a known sign. Conditioning on the intervening tokens is exactly what the forecast omits. Fixing S_u, the k-th term of Eq. (24) has expectation $H(Y_{u+k} \mid Y_{u+1:u+k-1}, S_u)$, whereas the MTP head models the *marginal* $p(Y_{u+k} \mid S_u)$. Since conditioning reduces entropy,

$$\begin{aligned} H(Y_{u+k} \mid S_u) - \mathbb{E}\big[H(\pi(\cdot \mid S_{u+k-1}))\big] \\ = I\big(Y_{u+k};\, Y_{u+1:u+k-1} \mid S_u\big) \\ \geq 0. \end{aligned} \quad (26)$$

The forecast *always over-estimates*, and the excess is precisely the mutual information with the intervening tokens, which grows monotonically with k.

Equation (26) is not merely a bound: it is the reason the marginal is the quantity we want. It is the fan-out of the continuation — how many ways the next k tokens can unfold — rather than the residual uncertainty along the one path that happened to be sampled.

B.4.3 A3: per-head affine calibration

Equation (26) predicts a per-head, monotone inflation, so we absorb it with a per-head affine correction fitted by least squares against oracle entropies on warm-up rollouts:

$$\hat{H}_k = a_k\, H\big(p^{(k)}_{\text{MTP}}(\cdot \mid h_u)\big) + b_k. \quad (27)$$

The fitted coefficients confirm the prediction. On our model the first head has $a_1 = 1.03$ — a one-step forecast has no intervening tokens, so the mutual information in Eq. (26) is zero and the approximation is exact — while a_k drops to 0.10–0.19 for $k \geq 2$. Calibration switches the unreliable heads off on its own.

Combining A1–A3 gives the forecast that is actually computed:

$$\hat{H}_{\text{togo}}(s_u \oplus y_u) = \max\Big(0,\; \sum_{k=1}^{\kappa} \gamma_H^k \big(a_k H_k(h_u) + b_k\big)\Big), \quad (28)$$

the clamp encoding the fact that an entropy cannot be negative. With $\kappa = 2$ and $\gamma_H = 0.7$ this instantiates to $\hat{H}_{\text{togo}} = 0.72\, H_1 + 0.06\, H_2 + 0.11$: the second head contributes about 8%, so the effective horizon is close to one step. We regard this as a consequence of Eq. (26) rather than a tuning artifact, and note it as a limitation — a forecaster whose error did not grow with k would leave a longer horizon usable.

Alignment. One detail decides whether the signal is a branch score at all. The hidden state at position u is conditioned on s_u and predicts y_u; it does not know y_u. The quantity we need, $H_{\text{togo}}(s_u \oplus y_u)$, must therefore be read from position $u+1$ and shifted back by one. Omitting the shift measures the value of the state *before* the branch rather than the value of the branch taken.

B.5 From Gradient to Token Weights

Equation (5) is a term to add to a gradient, but methods in this family do not add gradients — they re-weight a loss. What is needed is a first-order prediction of how much one update, acting through token u, changes the trajectory entropy.

How far one step moves a logit. We reuse the logit-space update of Hao et al. (2026, Eq. 23) rather than rederiving it. With $w_u = \text{clip}(r_u, 1 - \epsilon_{\text{lo}}, 1 + \epsilon_{\text{hi}})A_u$ the coefficient the policy loss actually applies, Assumption A makes a single step move only the sampled token's logit, $\Delta z_{u,a} = \eta\, w_u\, \delta_{a,y_u} + O(\eta^2)$. The softmax Jacobian $\partial \log \pi_a / \partial z_a = 1 - \pi_a$ then gives

$$\widehat{\Delta \log \pi}_u = \eta\, w_u\, (1 - \pi_u), \quad (29)$$

which vanishes as $\pi_u \to 1$: a token that already holds the mass cannot be given more. This is the

prefactor whose anti-correlation with $|\Omega|$ Table 1 records.

The visitation-channel prediction. Because $\nabla_\theta \log \pi_u \cdot \Delta\theta = \Delta \log \pi_u$, substituting into Eq. (5) gives the per-token quantity

$$\mathrm{visit}_u := \widehat{\Delta \log \pi}_u \cdot \mathrm{clip}\big(A_H(s_u, y_u), -c, +c\big). \tag{30}$$

A4 is the clip, and it is applied to A_H *before* the product rather than after, so that one token's influence is bounded by $|\eta w_u| \cdot c$. Clipping the product instead would let forecast error be amplified without bound at tokens with large w_u.

Sign quadrants. Equation (30) defines the collapse channel precisely. When $\widehat{\Delta \log \pi}_u > 0$ and $A_H > 0$, mass moves toward a branch richer than its siblings and trajectory entropy rises. When $\widehat{\Delta \log \pi}_u > 0$ and $A_H < 0$, mass moves into a dead end and the diversity carried by the abandoned siblings is destroyed — this is the collapse path that (L) is structurally unable to see. Negative $\widehat{\Delta \log \pi}_u$ reverses both readings.

B.6 Combining the Channels

The visitation term could be added to Ω before the weight map, but measured on real rollouts that map destroys the signal: Ω carries a factor $1/\pi_{\mathrm{old}}$, so $\max|\Omega|/\mathrm{median}|\Omega|$ reaches 3×10^4 and the band is spent on one or two outliers while the bulk lands on a point. We therefore apply the correction *after* the map, as a bounded adjustment of the weights the base method has already produced.

Let $\mathcal{S} = \{u : \mathrm{visit}_u \neq 0\}$ be the support — by Eq. (6), the branch points — and let $b = \lambda_{\max} - \lambda_{\min}$ be the width of the weight band. For $u \in \mathcal{S}$,

$$\alpha_u^{\mathrm{f}} = \mathrm{clamp}\Big(\alpha_u + \lambda\, b\, \tanh\big(\mathrm{visit}_u/\rho\big), \lambda_{\min},\, \lambda_{\max}\Big), \tag{31}$$

with $\rho = \mathrm{rms}\{\mathrm{visit}_v : v \in \mathcal{S}\}$; $\alpha_u^{\mathrm{f}} = \alpha_u$ elsewhere.

Three properties matter. The $\tanh$ is odd, monotone and bounded, so it preserves the sign of the visitation term, caps one token's movement at λb, and does not depend on the absolute scale of visit. The RMS is taken over $\mathcal{S}$ alone rather than over the whole response: normalizing by the whole response would shrink the correction in proportion to how sparse branch points are, which is a property of the sampler and not of the signal. And at $\lambda = 0$ the correction vanishes identically, so the pipeline is bitwise identical to the base method — a regression test enforces this.

Because the map runs first and the correction second, the step size η of Eq. (29) enters only through visit_u/ρ, where it cancels. We fix $\eta = 1$ and sweep λ alone.

The support is part of the treatment. $\mathcal{S}$ is defined by $\mathrm{visit}_u \neq 0$ rather than by a threshold, deliberately: it selects both where the correction lands and, through ρ, how large every correction is, so a tighter test would change what is trained rather than what is logged. One consequence is that $\mathcal{S}$ admits float32 round-off — A_H is a deviation from a mean over token-identical siblings, and $\sum_j g_j/m$ does not reproduce g_i bitwise, leaving $|A_H| \sim 6 \times 10^{-8}$ where it is interpretively zero. Those residuals are $\sim 10^{-6}$ relative to a real branch score and cannot move a weight, but they are counted in the support statistics of Appendix E.6, which should be read as an upper bound on the branch-point fraction.

B.7 Tree-Structured Rollouts

B.7.1 Why plain rollouts cannot support A_H

$\hat{H}_{\mathrm{togo}}$ is a deterministic function of the causal prefix, so two rollouts sharing $y_{\le u}$ produce identical forecasts. Combined with Eq. (5) this gives a sharp characterization:

$$A_H[i,u] \neq 0 \iff \begin{array}{l}\text{siblings share } y_{<u} \\ \text{but diverge at } y_u,\end{array} \tag{32}$$

that is, u is a branch point of the group. Under independent sampling from a sharpened policy, group members diverge at their first token and never share a prefix again, so $m = 1$ almost everywhere and $A_H \equiv 0$: the visitation channel is not merely weak, it is identically absent.

We therefore schedule the sharing. A tree sampler expands a single root and cuts it at fixed depths $d_1 < d_2 < d_3$ with branching factor 2, yielding $n = 8$ leaves whose sibling structure is known in advance. The cuts create the only positions at which Eq. (6) can fire. This is the smallest change to the sampler that lets the estimator exist, and it is all we ask of the tree: the depths are fixed rather than chosen, and we do not tune or claim the schedule as a way to sample.

B.7.2 A ceiling on the support

It is tempting to treat the fraction of positions with $A_H \neq 0$ as a quantity to be maximized. It is not,

and the reason is combinatorial rather than empirical. Two quantities behave very differently under the tree:

- the fraction of positions where *some* other rollout shares $y_{<u}$ is moved by the cuts, and equals d_L/T by construction;
- the fraction where $A_H \neq 0$ is *not*, because by Eq. (6) it counts branch points, and n sequences form a trie with at most $n-1$ internal branching nodes.

Since the one-position alignment shift widens each branch node into two columns, the fraction of positions carrying a non-zero A_H is bounded by $2(n-1)/T$ for *any* sampler whatsoever. With $n = 8$ and responses of $\sim 10^3$ tokens this ceiling is on the order of a percent. The tree's role is to make the branch points exist and to place them at controlled depths, not to make them numerous.

B.7.3 The correction cannot move the mean

A_H is a deviation from a sibling mean, so on each sibling set

$$\sum_{j \in \mathrm{sib}(s_u)} A_H\big(s_u, y_u^{(j)}\big) = 0. \tag{33}$$

The correction in Eq. (8) therefore cannot change the average weight assigned to branch tokens. It can only spread siblings apart. This is not a defect to be engineered away: an unbiased estimator requires a baseline, and a baseline forces the mean to zero.

The consequence for evaluation is direct, and we state it before presenting any experiment. "Mean entropy at branch points increased" is *not* a prediction of this correction and cannot serve as evidence that it acts; the corresponding prediction is that the *variance across siblings* widens. A correction designed to raise the mean is a different and weaker hypothesis — it does not need a forecast at all, only a support — and we include it below as a control arm.

C The Approximation Ledger

Table 2 collects every departure from Eq. (3).

The strongest assumptions are A2 and B2. A2 has a known sign and a known monotonicity (Eq. 26), which is what makes the affine correction in A3 an appropriate remedy rather than a curve fit. B2 — that updating one token does not change the conditional distribution at other positions — is false for a network with shared parameters and has no corrective mechanism. It is not a risk this work introduces, since the base method already relies on it, but adding a second channel adds a second copy of its error.

D Experimental Details

D.1 Model, data, and optimization

All runs fine-tune the same base model, Qwen2.5-Math-1.5B (Yang et al., 2024), on the same data with the same optimizer settings, and differ only in the intervention; the five-arm decomposition and every ablation are run on it. We run no second backbone, and the Limitations say what that leaves open. All runs train on the same 17k-problem competition-mathematics set, DAPO-Math-17k (Yu et al., 2025). With a training batch of 512 prompts, 110 optimization steps correspond to about 3.2 epochs. Two further runs of the SIGNED configuration at 200 steps, under the base method's own protocol, are described separately in Appendix F. Licenses: Qwen2.5-Math-1.5B, DAPO-Math-17k and verl are released under Apache-2.0; MATH (from which MATH500 is drawn), GSM8K and OlympiadBench under MIT; the Minerva evaluation problems come from MIT OpenCourseWare, released under CC BY-NC-SA. AIME and AMC problems are publicly released competition problems. Every artifact is used for non-commercial research on mathematical reasoning, its intended use.

D.2 Validation set

Every run validates on AIME 2024 (30 problems, 32 samples each) every 10 steps. AIME 2024 is small: with a per-problem accuracy spread of .1547 measured across our runs, the standard error of its mean is $.1547/\sqrt{30} \approx .0282$. That figure governs two things. It bounds what any single validation point can resolve, which is why the primary statistic below is a window mean rather than one point; and it bounds checkpoint selection, since a benchmark this noisy chooses between nearby checkpoints partly by chance. Where a checkpoint is selected (the six-benchmark evaluation of Appendix E.9), the AIME24 column is therefore marked in-sample.

Optimization uses GRPO (Shao et al., 2024) with $n = 8$ responses per prompt, a constant learning rate of $1\mathrm{e}{-6}$ with no warm-up (so that the schedule is independent of the declared total step

count), a maximum response length of 3072 tokens, no entropy bonus, and no KL loss. The clipped surrogate (Schulman et al., 2017b) is asymmetric, $\varepsilon_{\text{low}} = 0.2$ and $\varepsilon_{\text{high}} = 0.28$ as in Yu et al. (2025), with the dual-clip lower bound at $c = 10$; we state this because "GRPO" with a symmetric $\varepsilon = 0.2$ and a KL term is a different baseline, and because the follow-up arm of Appendix E.10 is defined by moving these two numbers. Every arm, the baseline included, uses the same three values: the only configuration difference between the GRPO arm and SIGNED is the loss mode and the correction hyperparameters listed next. Rollouts are sampled at temperature 1.0 with no nucleus truncation; validation uses temperature 1.0 and top-$p = 0.7$. Every arm uses the same data seed and the same initial checkpoint, which we verify by requiring the step-0 validation scores to agree.

D.3 Correction hyperparameters

The forecaster uses $\kappa = 2$ heads with discount $\gamma_H = 0.7$ and the per-head calibration of Eq. (27). The combination uses $\lambda = 0.25$, RMS normalization, clip $c = 1.0$, and the sibling-prefix baseline. The weight map is the base method's exponential map of Eq. (10), taken un-rescaled, at $\lambda_{\min} = 0.7$, so weights land in $[0.7, 1]$ and can only attenuate. This is the value the base method's paper sets, inside the range $[0.6, 0.8]$ over which it reports the method performing consistently (Hao et al., 2026, App. F.1); it was taken from there and not tuned on these runs. The base method's released launcher uses $\lambda_{\min} = 0.8$, and every other value in that launcher matches ours.[1] The tree sampler uses a single root, cuts at depths $\{64, 192, 384\}$, and branching factor 2 at each cut, giving $n = 8$ leaves.

D.4 Five arms

The design isolates one question: does the *value* of the forecast matter, or only the apparatus that carries it? A comparison between the base method and SIGNED changes two things at once (λ and the sampler), so it cannot answer this. We therefore add two arms that keep the entire apparatus — the tree, the forecaster, the support, the weight mapping — and destroy only the information in A_H.

UNIFORM applies the same attenuation to every position in the support, discarding A_H's value but keeping its support — this is the implementation of the "raise the mean" hypothesis that Section 4 distinguishes from ours. PERMUTED keeps the multiset of A_H values within each sibling set and shuffles the assignment, so the support, the magnitudes, and Eq. (33) all survive while the pairing between a branch and its score is destroyed. Together they bracket the claim: if the visitation signal is doing the work, both controls should behave like the no-forecast arms.

D.5 Metrics

We report acc@32 (mean accuracy over 32 samples) and maj@32 (accuracy of the majority vote over the same 32 samples), and their difference

$$\text{uplift} := \text{maj@32} - \text{acc@32},$$

which measures how much better a set of 32 samples is than its average member — that is, whether correct answers concentrate while incorrect ones scatter. We use uplift rather than pass@32 because the latter does not separate the arms.

Because single validation points are noisy on a 30-problem set, the primary statistic is the mean over the converged window, steps 40–110 (8 validation points per arm). We quote differences in units of the across-problem standard error, $\text{SE} = \text{std@32}/\sqrt{30} \approx 0.027$, which is the correct scale for generalization to new problems. We separately note that the pure re-sampling noise of the protocol — estimated from two validations of identical weights — is about 0.004 in accuracy, an order of magnitude smaller; the two scales answer different questions and we are explicit about which is in use.

One instrumentation caveat propagates into every entropy number we report. The logged `actor/entropy` is aggregated with a token-mean, so it is $\mathcal{H}$ divided by the mean response length, not $\mathcal{H}$. Since response lengths differ by under 5% across arms, we report the raw quantity in the main tables and give the length-corrected quantity for the dissociation contrasts in Table 9. It is computed at every step as the token-mean entropy times the mean response length, then averaged over the same validation steps as every other entry. The runs on seeds 3 and 4 also log the per-response sum directly (`actor/seq_entropy`); on all 1,100 steps of those ten runs the product agrees with it to within 0.5%, the rounding of the logged token mean. The seed-1 runs predate that key, so the product is used throughout.

[1] Our implementation exposes an optional reshaping of $\tilde{\Omega}$ before this map (winsorizing or rank-transforming it); every run reported here uses the identity setting, so the base method's map sees exactly what it would have seen.

contrast	n	$\Delta\mathcal{H}$ (nats)	t	seeds < 0
the damping mechanism changes, the A_H signal does not				
STEER − GRPO	3	−42.5	−4.47	3
UNIF − STEER	3	+16.6	+4.43	0
the A_H signal changes, the damping mechanism does not				
SIGNED − UNIF	3	−6.4	−1.39	2
SIGNED − PERM	3	+1.8	+0.99	1
both change				
SIGNED − STEER	3	+10.2	+1.73	1
SIGNED − GRPO	3	−32.3	−6.67	3

Table 9: The contrasts of Table 4 on the length-corrected trajectory entropy $\mathcal{H}$ (token-mean entropy × mean response length, nats per response), paired within a seed. The mechanism contrasts move $\mathcal{H}$ in the same direction on every seed; the signal contrasts do not.

E Additional Results

All numbers below come from the runs described in Section 5: 110 optimization steps per arm from the same initial checkpoint, validated on AIME24. Every figure is recomputed by one script from a training log included in the supplementary material; we report our own measurements only, and no figure in this section is taken from prior work. Appendix F, which follows it, is the one place that does.

Every statistic is over the same 3 seeds, $1, 3, 4$ — the seeds on which all five arms finished — so a difference of two rows of Table 5 is the corresponding paired entry of Table 10. Seed 2 is excluded because its SIGNED run did not complete; its other four arms did, and every run, excluded or not, is in the `per_seed.tsv` of the supplementary material. Appendix E.8 reports what the repeats measure — the run-to-run spread, and the machine each seed ran on.

Which result carries which weight. What follows is not one kind of result, and the order is the order of how much each one bears. Section 6.1 comes first. It tests a prediction the construction makes before any run — that a correction this sparse and this mean-preserving cannot move an average taken over every response token — and it tests it as a contrast *between groups of contrasts*, on 3 seeds, with the two groups' $|t|$ not overlapping. That is the structural claim, and it is what this campaign establishes. The accuracy contrasts after it rest on 3 paired seeds each, and they do not all survive: the gap against STEER clears a conventional threshold, the gap against PERMUTED is marginal, and the headline gap against GRPO is smaller than GRPO's own spread across its seeds (Appendix E.8), does not, and is close to zero on the seeds where equal steps mean equal compute (Appendix E.11). One pre-registered falsification condition is met at this seed count, on the contrast that isolates A_H's value; Section 6.3 states which and what it leaves standing. A reader who wants one result from this section should take the dissociation, not the gap.

E.1 Validation trajectory

Table 6 gives AIME24 accuracy and majority-vote accuracy at every validation point of seed 1, the one seed every arm ran first; the other seeds' trajectories are in the supplementary logs. Four of the five arms start from an identical step-0 score; STEER starts at .043 rather than .039, a gap we attribute to resampling noise rather than to a different checkpoint, since two validations of the *same* base weights in our logs differ by .004 in accuracy.

E.2 Converged window

Table 5 averages each arm over steps 40–110.

Two orderings are worth stating before anything is subtracted. On accuracy all three arms that do not carry A_H's value — STEER, UNIFORM, PERMUTED — sit *below* plain GRPO, and SIGNED is the only one above it. It is above by less than GRPO's own standard error across seeds, which is why the ordering and not the size is what we take from this table. On entropy GRPO holds the most by a clear margin and the four intervened arms sit close to one another, SIGNED among them, so the two columns are neither aligned nor reversed. Neither ordering is the instrument for either question at this seed count, and the next section says why: read as contrasts rather than as a ranking, the entropy column tracks the damping apparatus and the accuracy column tracks A_H.

E.3 Entropy: the aggregate moves with the apparatus, not with A_H

What the analysis predicts here, fixed in advance. The correction does not set out to hold the aggregate entropy up, and two facts already established above say it cannot. Appendix B.7 bounds the positions at which A_H can be non-zero at $2(n-1)/T$, about one in eighty; Section 4 shows A_H sums to zero over a sibling set, so at the positions it does reach it redistributes weight among siblings rather than adding any. An intervention that is this sparse and mean-preserving by construction should leave an average over every response

token close to where it found it. Both statements were written into the manuscript before the campaign was launched, Section 4 explicitly so ("we state it before presenting any experiment"), and the repository history, to be made public with the code, dates them an hour ahead of the pre-registration discussed below.

What the seeds show. Table 4 groups the contrasts by what each one changes, and the grouping separates them cleanly. The two contrasts that change the apparatus move the aggregate: turning the damping on costs $-.0419$ at $t = -5.87$, and adding the tree with damping applied uniformly returns $+.0184$ at $t = +13.09$ — the latter reproducing across all 3 seeds to within a thousandth. The contrasts that hold the apparatus fixed and change only what A_H contains move it by $-.0077$ ($t = -1.42$) when its value is discarded and $+.0044$ when its pairing is destroyed. Accuracy runs the other way: the apparatus contrasts give $+.0051$ and $-.0065$, while the two A_H contrasts give $+.0069$ and $+.0087$. The quantity that moves the aggregate entropy is the damping apparatus; the quantity that moves accuracy is A_H. That is the predicted pattern, and it is the sense in which this paper is not a method for suppressing entropy.

A pre-registered condition we retire rather than score. The pre-registration (Appendix I) also fixed a condition on this section: if aggregate entropy orders the arms the same way accuracy does, a dissociation argument fails. We retire that condition instead of reporting it, and say plainly why, since retiring a registered test is exactly the move a reader should be suspicious of. First, it tests the aggregate as a signal about A_H, which the two paragraphs above — and Section 4, committed an hour before the registration was — say it cannot be. Second, an ordering over five arm means is not stable at this seed count: the rank correlation between converged entropy and converged accuracy comes out with a different sign depending on whether each arm is averaged over the seed common to all five (Table 5) or over its own, which differ in number while the campaign runs. Both seed sets are in the supplementary `per_seed.tsv`. Third, and because of the first two, *we claim nothing from it in either direction*: the condition is not reported as passed, and the theory is not reported as damaged by it. It was the wrong instrument to have registered, and the honest correction is to say so and to register a better one.

The measurement that is actually owed. Section 4 states the prediction that a sum-zero correction does make: not that entropy at branch points rises, but that the *spread across siblings* widens. No quantity we currently log tests it directly. The trainer does log an entropy split (`steerf/branch_entropy`), and an earlier draft of this section read it as entropy at branch points; it is not. That monitor takes a fixed top decile of A_H — a tenth of all response tokens — while the positions A_H can reach are the one-in-eighty that Appendix E.6 measures, an eighth as many. Since A_H is exactly zero at almost every position, the remainder of that decile is filled by an arbitrary tie-break rather than by branching. Appendix E.6 reports what that monitor does and does not select. The replacement restricts the split to $A_H \neq 0$, the same predicate the correction itself uses, and is instrumented in our trainer; no run reported here carries it, so this test is registered and not performed. That test is registered in Appendix I, before those runs, and we state here that it is registered after this section's original condition was retired rather than before.

What the numbers above cannot carry. The claim is about magnitudes, not about one row failing a threshold. Changing the apparatus moves the aggregate by $+.0184$ and $-.0419$; changing what A_H contains moves it by $+.0044$ and $-.0077$, and the smaller of the two apparatus effects is still more than twice the larger of the two A_H effects. The $|t|$ bands separate the same way and do not overlap — -5.87 and $+13.09$ against -1.42 and $+2.15$ — which is why the argument is the contrast *between groups* rather than any single row. Two cautions travel with it. A small difference that reproduces across seeds can still reach a large t: the A_H contrast on PERMUTED does, at $+.0044$, and that is a statement about consistency, not about size. And 3 seeds entitle us to say the aggregate does not move *much* when A_H changes, not that it is unchanged.

E.4 Does the forecast's value matter?

Table 10 averages each arm over its converged window, pairs the arms within a seed, and reports the mean difference together with a paired t on those seed-level differences. The step-by-step version of this statistic, which the single-seed draft reported, is in Appendix J; it answers a question about one trajectory rather than about re-running.

contrast	n	Δacc	t	Δmaj	t
SIGNED − GRPO	3	+.0055	+0.81	+.0155	+1.34
SIGNED − STEER	3	+.0120	+5.24	+.0211	+2.37
SIGNED − UNIF	3	+.0069	+1.16	+.0187	+1.77
SIGNED − PERM	3	+.0087	+3.11	+.0172	+2.67
STEER − GRPO	3	−.0065	−1.24	−.0055	−0.87
UNIF − STEER	3	+.0051	+1.37	+.0024	+0.39

Table 10: Contrasts paired *by seed*: each arm is averaged over its converged window, arms are paired within a seed, and t is the paired t on the n seed-level differences ($\mathrm{df} = n-1$). Pairing within a seed cancels any per-seed effect common to all five arms. A contrast's n is the smaller of its two arms' seed counts.

These contrasts carry less than the dissociation above, and we read them for direction rather than magnitude. They pair on the same 3 seeds as that section, and the quantity they measure — an accuracy gap — is the one the seeds in hand do not separate from run-to-run spread (Appendix E.8).

The block structure is what the design was built to expose. Every contrast that changes the *value* of A_H is positive; the one that leaves it alone is not distinguishable from zero. UNIFORM runs the entire apparatus — tree sampler, forecaster, support, weight map — and differs from STEER by $+.0051$ ($t = +1.37$ over 3 seeds): turning the machinery on and feeding it a constant buys nothing. PERMUTED keeps the machinery *and* the magnitudes of A_H and destroys only the pairing between a branch and its score, and SIGNED still exceeds it by $+.0087$.

Every contrast here now rests on 3 paired seeds, and they do not all say the same thing. Against STEER the gap is $+.0120$ at $p = 0.035$; against PERMUTED it is $+.0087$ at $p = 0.090$. Against UNIFORM and against GRPO it is 0.366 and 0.503 — the headline contrast is the weakest row in the table, for the reason Appendix E.8 gives: it is the one that changes three things at once, and GRPO's own spread across seeds is wider than the effect. The signs agree in the mean everywhere A_H carries information, but the per-seed signs do not, and the next paragraph is about the one place they part.

A pre-registered falsification condition is met. The pre-registration (Appendix I) named four things that would count against this paper, before any of these runs existed. One of them has happened, and it is the one that clause calls the claim of the paper: *if* SIGNED − UNIFORM *straddles zero across seeds, the forecast's value does not matter and the apparatus explains the effect.* On accuracy it straddles. Across the 3 paired seeds the difference runs from $-.0020$ to $+.0182$, with 1 of them negative. We report that as a refutation of the registered claim rather than as noise, and we do not rescue it with the mean or with the majority-vote column, where the same contrast happens not to straddle (0 of 3 negative) — the registered primary endpoint is converged-window accuracy.

The other three conditions did not trigger, and the contrast between the first and the second is the informative part. SIGNED − PERMUTED, which holds the magnitudes of A_H and destroys only the pairing, does *not* straddle: $+.0033$ to $+.0125$, 0 negative. Aggregate entropy does not order the arms the way accuracy does — SIGNED is third from the top on one column and first on the other in Table 5 — so the dissociation argument of Section 6.1 stands. And the mean token weight stays at .9990, so A_H is sum-zero over sibling sets as derived.

Read together: the *pairing* of A_H to a branch survives its registered test and its *value* does not. The two arms were built to separate exactly those two things, so this is the distinction the design can draw, not an evasion of the result. It leaves the paper with a tension we do not resolve here. The pre-registration calls the UNIFORM contrast the central claim; the argument this paper now leads with is the dissociation of Section 6.1, which on these same 3 seed pairs separates entropy from accuracy at $p = 0.005$ against $p = 0.503$. Both are on the record; a reader who weights the pre-registration above the reframing should read the accuracy claim as refuted at this seed count.

The same contrast on questions the checkpoints were not chosen on. The registered secondary check of this reading is the six-benchmark evaluation of Appendix E.9. On the evaluated seed (1) it counts SIGNED above UNIFORM on 3 of six benchmarks, which the pre-registered thresholds call *equivocal*, and on 2 of the 5 that played no part in selection; paired over 1445 problems, the difference is -0.0090 ± 0.0099 ($t = -0.91$). It does not change the condition above, which is defined on converged-window accuracy across the training seeds. It is the registered check on other questions, reported because it was registered and not as a rescue.

This is the contrast the design was built for. A comparison of SIGNED against STEER alone changes two things at once, and cannot separate

the forecast from the sampler that delivers it; the two destroyed-forecast arms do separate them.

E.5 Where the gain comes from

The majority-vote gain decomposes into a part explained by per-sample accuracy and a part explained by the sample set's structure:

$$\underbrace{+.0155}_{\Delta\text{maj}} = \underbrace{+.0055}_{\Delta\text{acc}} + \underbrace{+.0100}_{\Delta\text{uplift}},$$

against GRPO; against STEER the uplift term is $+.0091$ ($t = +1.24$ over 3 seeds). Against GRPO roughly two thirds of the improvement is not that individual samples are more often right, but that the right ones agree with each other more and the wrong ones agree less. Both terms are means over the same 3 seeds, and the split inherits the weakness of the gap it splits: against STEER the uplift term is smaller and well inside its own spread, and against GRPO the whole gap is inside GRPO's.

Where that structural change comes from is a separate question, and the arm that answers it finished after this section was first written. UNIFORM runs the sampler at $\lambda = .25$ with the forecast's value destroyed and shows no uplift gain, which reads as though the structure belonged to A_H. It does not: the $\lambda = 0$ tree arm — the sampler with the future channel off, STEER's local damping still in place — reaches an uplift of .0870, against .0719 for STEER and .0810 for SIGNED. On this evidence most of the structural gain is the sampler's, and Appendix E.10 reports it that way. The decomposition above is unaffected — it is measured against GRPO, which has no tree — but the attribution is not what the five-arm design alone could see. A gain that belongs to the sampler is outside what this paper set out to study, since the tree is here only to give A_H its siblings; we report it and leave its explanation to work that studies structured sampling directly.

E.6 Mechanism diagnostics

Appendix B.7 bounds the fraction of positions that can carry a non-zero A_H at $2(n-1)/T$ for any sampler. With $n = 8$ and the tree arms' mean response length of 977 tokens over the converged window the bound is .0143. Measured over the same window and seeds, the three tree arms sit at .0131 (UNIFORM), .0123 (PERMUTED) and .0125 (SIGNED), i.e. at 86–91% of a ceiling that no sampler can exceed. A plain-rollout pilot with the same λ (seed 1, included with the supplementary logs) reaches .003. As Appendix B.6 notes, this count admits float32 round-off and is an upper bound; the runs reported here predate the round-off-free counter, so we quote the loose figure and label it as such.

Two readings follow. The tree sampler is close to saturating what the combinatorics allow, so there is little headroom in scheduling more cuts; and the intervention is extremely sparse — about one position in eighty — which makes the size of the effect in Table 10 the surprising part rather than the small support.

That sparsity also disqualifies one of our own monitors, which we record here rather than leave for a reader to find. The logged entropy split (`steerf/branch_entropy` and its gap) selects the top decile of A_H by rank, a tenth of all response tokens, against a support an eighth that size. A rank-based cut was chosen deliberately — A_H is exactly zero wherever a rollout has become its own only sibling, so a value-based threshold selects almost everything — but with a support this small the decile is filled out by the tie-break among those zeros, and the tie-break is not a random sample of them. A top-decile cut does recall about half of the true divergence points, five times what selecting at random would give, so the bucket is genuinely enriched; it is nonetheless mostly not branch positions, and the mean entropy it reports cannot be attributed to them. Section 6.1 states what replaces it.

E.7 Is the future term a local term?

Appendix B.4 reports a calibration that suppresses every head past the first, leaving $\hat{H}_{\text{togo}}$ dominated by one step ahead. That invites the sharpest objection to the paper: if the forecast reaches one step, is A_H anything a purely local quantity could not have produced? The question is answerable without training anything. We apply the *same* operator — the same sibling baseline, mask and grouping — to two inputs: the calibrated forecast, and the policy's own next-token entropy read at the same offset, the state after the branch token. Any difference between the two is the forecaster's contribution and nothing else. We report it on the sibling support, where A_H can be non-zero at all; outside it both are identically zero and a statistic over all positions measures the support rather than the forecast.

The answer is that the forecaster is not earning its cost. Over 464 branch points from 4 val-

idation prompts, the two advantages correlate at $r = +0.917$ (Spearman $+0.937$), the local twin explains $R^2 = 0.840$ of the treatment, and the two agree in sign 0.921 of the time — and the sign is what the token weights are monotone in. The levels correlate less than the advantages ($+0.881$), which is the wrong way round for an artefact of scale.

The mechanism is one the paper had already measured without connecting it. With $\kappa = 2$ and $\gamma_H = 0.7$ the calibration instantiates to $\hat{H}_{\text{togo}} = 0.72\,H_1 + 0.06\,H_2 + 0.11$, the sibling baseline removes the constant, and the first head is built to reproduce the policy's own next-token distribution — its final layer is zero-initialised and its fitted $a_1 = 1.03$ says it stayed there. So A_H should be close to $\gamma_H a_1 = 0.72$ times the local twin, and the measured magnitudes are 0.3445 against 0.4647, a ratio of 0.741. A_H is very nearly a scalar multiple of a quantity the trainer already computes, and Eq. (8) divides the scale out through $\rho = \text{rms}\{\text{visit}\}$.

Two things this does *not* say. It does not touch Eq. (3), which is an identity, nor Section 3.1: the local twin is not Ω. Ω carries no sibling baseline and no tree, vanishes at a uniform branch point, and the twin exists only because the tree supplies siblings to difference against. What the measurement indicts is the *forecasting apparatus* — the MTP heads, their warm-up and their calibration — not the visitation channel. Read alongside Appendix E.10, where the sampler alone recovers .1439 against STEER's .1371 while the forecast adds the remainder, two independent measurements now point the same way.

The caveats are the ones a single diagnostic run carries: 4 prompts at one validation step of the base policy, on independently sampled rollouts, whose sibling support therefore sits early in the response rather than at the scheduled branch depths. The head-free arm of Table 7 is the direct test, and it has now been run. ORACLE reads H_{togo} from the policy's realised entropy, which verl (Sheng et al., 2025) computes in the same pass, so it carries none of the MTP overhead of Appendix E.12 — 416 s per optimization step against SIGNED's 516 at the same seed on the four-GPU box, 19.3% less. It is also behind it on accuracy. Appendix E.10 reports both numbers and what one seed of each can carry.

E.8 Seeds, not steps

Every statistic above treats the seed as the unit of analysis: each run is first collapsed to one number by averaging its converged window, the arms are then paired within a seed, and the test is a one-sample t on the paired differences with $\text{df} = n - 1$. Pairing within a seed matters because a seed carries effects common to all five arms — the data order, and on a split campaign the machine a run landed on — and those cancel exactly in a difference. A contrast's n is the smaller of its two arms' seed counts, which is why the table reports n per row rather than once.

The alternative statistic, a paired t over the eight window steps *within* one run, measures whether a gap is stable along a single trajectory. That is a different question from whether it survives re-running, and we report it only in Appendix J so the two are not confused.

Mechanical failures, and what replaced them. The pre-registration (Appendix I) excludes a run for mechanical failure and relaunches it with the same seed. SIGNED at seed 4 stopped at training step 91 when its machine lost a GPU; it was relaunched at seed 4 and the relaunch, which reached step 110, is the run reported. SIGNED at seed 2 failed at start-up on a missing module and was not relaunched within this campaign's budget, which is why seed 2 is outside the seed set. Both failed runs are in the supplementary logs, flagged rather than deleted.

A seed index also names a machine. A seed of this campaign ran on a two-GPU box or a four-GPU one, uniformly across arms, and the seed set this paper uses — $1, 3, 4$ — holds 1 of the first and 2 of the second for every arm. The effect is measurable on $GRPO$, which has both: .1350 on the smaller box against .1479 on the larger, a difference of $+.0129$, larger than the gap between the top two rows of Table 5.

Read that difference as an upper bound and not as an estimate. With 1 run on the smaller box it is one run against 2, and a single run carries this arm's own run-to-run spread (Appendix E.8 below) inside it. We report it at that strength because the alternative was to keep a seed outside $1, 3, 4$ alive for this one number while every other number in the paper excluded it, and a nuisance estimate is not worth a second seed set.

What the figure does not reach is the comparisons, for two reasons that do not depend on how well it is estimated. The contrasts pair inside a seed and therefore inside a machine, so whatever a box is worth cancels in the pair. And every arm in Table 5 carries the same 1-to-2 mix, so a row

difference does not carry a box mix either. Our analysis prints the per-arm machine mix on every run.

Run-to-run spread, and it is the number that governs this section. Every arm has now finished more than once. GRPO's converged-window accuracy moves from .1350 to .1560 across its 3 seeds, a spread of .0210; SIGNED moves from .1481 to .1497 across its 3 seeds, a spread of .0016; PERMUTED moves only .0095 across 3. Both extremes are measured on the same benchmark with the same protocol, and they differ by more than an order of magnitude.

Set the headline gap of +.0055 against those. **It is a fraction of GRPO's own spread across seeds and several times SIGNED's**, so the two arms it compares do not even agree on how large a seed-to-seed difference is, and each is observed a handful of times. That is the regime in which a single paired difference cannot decide the question, and we do not claim it does. A range between three draws is also not a standard error: the largest of these ranges is more than ten times the smallest, so they do not bound each other and no contrast here should be read against a noise floor quoted to one significant figure. For scale at the other end, two validations of the *same* weights on these 30 questions differ by about .004, so seed variation is the larger term by roughly an order of magnitude and any comparison quoted at single-run precision is quoting noise.

We therefore state what this experiment establishes and what it does not. The mechanism contrasts — what changes when A_H's value is discarded or its pairing destroyed, with the apparatus held fixed — are the claim. The effect size would need more seeds than this campaign has, and on the seeds it has the headline gap is not separated from the spread.

E.9 Six benchmarks

A single 30-question benchmark cannot separate a method from a lucky draw, so each run's selected checkpoint is re-evaluated on six mathematics benchmarks (Table 11) under the protocol of Section 5: AIME24, AIME25 and AMC23 at avg@32, MATH500 (Hendrycks et al., 2021; Lightman et al., 2024), Minerva (Lewkowycz et al., 2022) and OlympiadBench (He et al., 2024) at avg@1.

An evaluation enters the table only if its log agrees with itself. The trainer prints each benchmark twice on the same line, as accuracy and as the mean of a reward that is $+1$ for a correct sample and -1 otherwise, so on every benchmark the accuracy must equal half of one plus the reward; our analysis checks this for every evaluation log, and the supplementary logs let a reader repeat the check. It is a test of coherence, not of provenance — an edit that moved both numbers together would pass it. Absent from the table, for want of an evaluation that passes it: STEER. Each row averages the checkpoints evaluated at the seeds in its last column.

arm	AIME24†	AIME25	AMC23	MATH500	Minerva	Olympiad	mean	seed
GRPO	15.8	8.2	54.8	71.4	26.5	33.9	35.1	3
UNIFORM	15.7	6.7	56.0	72.2	25.7	33.6	35.0	4
PERMUTED	15.2	6.4	55.8	69.0	23.9	31.2	33.6	3
SIGNED	17.1	7.3	52.5	69.4	23.9	34.5	34.1	4
ORACLE	14.2	8.2	58.2	70.8	24.3	33.7	34.9	4

Table 11: Six-benchmark evaluation of each arm's selected checkpoint, averaged over the seeds in the last column. The mean column is the unweighted mean over the six. †: AIME24 is the set each checkpoint was selected on (Appendix D.2), so that column is in-sample; the other five are held out. Rows at different seeds are not meant to be subtracted — a seed index also names a training run and the machine it ran on — and the contrasts below are counted within a seed.

One of the six is not held out. The checkpoint behind each row is the argmax of that run's AIME24 validation accuracy (Appendix D.2), so AIME24 is the set the selection was made on and its column is in-sample: the pre-registration (Appendix I) records that this argmax lifts an arm by about a hundredth in absolute accuracy, and by an amount that differs between arms, so the bias does not cancel cleanly inside a contrast. The column is marked † and kept — dropping a registered endpoint after seeing it is the worse of the two errors — and the five remaining benchmarks, which played no part in selection, carry no such bias and only more variance.

The quantity fixed in advance as the headline here is not any single cell but *direction consistency*: in how many of the six benchmarks the ordering observed on AIME24 is preserved. Four or better of six we declared support, three equivocal, two or fewer evidence against. The observed count is 3 of six for SIGNED $>$ UNIFORM, which those thresholds call *equivocal*, over the seeds the two arms share (1). The SIGNED $>$ PERMUTED count is not reported: a count needs the two arms evaluated at a common seed, and the evaluations that passed the

coherence check do not include one. Counting the benchmarks on the 5 that played no part in selection gives 2: the registered six-count is the decision rule, and the held-out count is printed beside it so a reader can see how much of it survives without the set the checkpoints were chosen on. This is the test that distinguishes a method from an artifact of one question set, and it is reported whatever it returns.

These two contrasts carry the attribution because each changes one thing: UNIFORM has the apparatus without A_H's value, PERMUTED has its value without its pairing. The differences against GRPO and STEER are registered as well and appear in Table 10, but they change three things and two respectively, so a benchmark difference there names no cause. Arms are paired inside a seed throughout, for the reason Table 5's caption gives; the arm-mean rows of Table 11 are not what is being counted.

The paired error bar. A single measurement of one arm on AIME24's 30 questions carries an across-problem standard error of .0282, which is why the gaps in Table 10 are a fraction of one problem in absolute terms. That figure is the wrong one for a contrast: the two arms answer the same questions, so per-problem difficulty cancels between them and what matters is the standard error of the *paired* difference. This evaluation records one row per generated sample, so on the three avg@1 benchmarks with enough problems to support it — MATH500, OlympiadBench and Minerva, 1445 problems together, over the seeds the arms share (1) — the paired quantity is computed rather than asserted. SIGNED − UNIFORM is -0.0090 ± 0.0099 ($t = -0.91$) against an unpaired error of 0.0185; the PERMUTED contrast has no common seed and so no paired figure. How much pairing buys is decided by how often the two arms disagree on a problem, which is 0.142 here; the measurement is what we report, not the reduction we wanted. Two cautions travel with it. At avg@1 each problem's score is a single Bernoulli draw, so these errors carry the sampling noise of one generation per problem on top of the difficulty spread; and pooling problems across seeds treats a (seed, problem) pair as one observation, which ignores the clustering by seed that Appendix E.8 shows is real. GSM8K is computed the same way and left out of the pool: at avg@1 it sits near saturation, where a paired difference measures the ceiling rather than the correction.

E.10 How much of the design is load-bearing

Table 7 varies one element at a time, at a single seed per row. Every comparison in it is therefore $n = 1$: these rows describe a mechanism, and none of them carries an error bar or licenses an effect size. A row is read against the main arm at *its own* seed, which is not the same number as Table 5's across-seed mean.

The coefficient. λ scales the visitation term before the two channels are combined in Eq. (8). The value used throughout was not tuned on these runs, and the sweep exists to say whether the result rests on a knife edge. A flat response across 0.1–0.5 would say the correction is insensitive to it; a peak at 0.25 with both neighbours below would say it is a tuned quantity and must be reported as one.

The base method against GRPO, and what it does and does not reach. We measure STEER below GRPO by $-.0065$ over 3 seeds, where the base method's own results place it above. We read the released weighting function against ours and it agrees line for line, as does every other value in the two launchers except $\lambda_{\min}$, which we set to the paper's 0.7 where the released script uses 0.8. That difference does not explain the gap on the base method's own evidence: both values lie in the range its sensitivity analysis reports as performing consistently. The settings that do differ from the base method's main experiments are the schedule (110 steps against its 200), the checkpoint rule (a fixed window against its best checkpoint) and the number of runs, and we have no measurement that separates them. We did not run the base method at the released 0.8, so the question of why it lands below GRPO here is left open rather than attributed.

What this discrepancy can reach is bounded, and by the design rather than by luck. All four damped arms share $\lambda_{\min} = 0.7$ exactly, so the setting cancels in any contrast between two of them. SIGNED − UNIFORM and SIGNED − PERMUTED — the pair that carries this paper's central claim, that the forecast's *value* is what matters — hold $\lambda_{\min}$, λ, the sampler and the weight map all fixed and vary only what A_H contains. Those two do not depend on whatever puts the base method below GRPO here. What it does reach is how two other contrasts read. STEER − GRPO is the discrepancy itself. SIGNED − STEER is measured against a base method that is not showing its reported gain over GRPO, so it is

not a measurement of improvement on top of that gain. We report both with this stated rather than resolved.

The sampler. This is the cell no other arm supplies. Tree rollouts make a group share prefixes, which changes the GRPO advantage estimate itself, quite apart from any damping. UNIFORM and PERMUTED both run at $\lambda = .25$ and so cannot separate the sampler from the correction. The $\lambda = 0$ tree arm switches the forecast off while leaving the sampler in place, and is the direct answer to "is tree rollout simply a better sampler". Read against STEER, which has neither, its difference is the sampler's own contribution; read against SIGNED, what remains is the forecast's.

The answer is not the one the rest of this paper was written expecting. The sampler alone reaches .1439 against STEER's .1371, and .2309 against .2090 — a little over half of SIGNED's accuracy margin over STEER and all of its majority-vote margin. Its uplift, .0870, is not below SIGNED's .0810. What remains for the forecast is the accuracy gap between .1439 and .1491.

Three things keep this from being decisive, and none of them rescues the original reading. Every cell here is one run, and the spread across this block is about twice the run-to-run spread measured in Appendix E.8 — suggestive, not settled. The λ sweep is not monotone: $\lambda = .1$ sits below $\lambda = 0$ on accuracy, which is what noise looks like at this sample size. And the row ran on the 4-GPU box while the seed-1 references it is read against did not, so the $+.0129$ of Appendix E.8 sits inside this comparison and points the same way as the conclusion drawn from it: the sampler's share is an upper bound, not an estimate. We report the block as measured and do not reconcile it. Whether the sampler or the forecast carries the structural term needs these arms at more seeds, which this paper does not have. Nothing here establishes that the tree does not carry it, and Section 6.3 is explicit that the forecast's *value* does not survive its own registered test. We do not turn this cell into a claim about the sampler either: the schedule was fixed in advance to give A_H its siblings, it was not designed or tuned as a sampler, and whether structured sampling improves RLVR in its own right is a different question from the one this paper asks.

The forecaster, replaced by what the trainer already has. Appendix E.7 builds the same sibling-differenced quantity from the policy's own next-token entropy and finds it close to A_H. That measurement is taken on a frozen policy at one validation step, and it cannot say what the substitution does *inside* the loop, where the weights it produces move the trajectory the next measurement would be taken on. The ORACLE arm makes the substitution there: same tree, same λ, same weight map, with H_{togo} read from the realised entropy of the rollout verl has already computed. A2 and A3 — the head substitution and the affine calibration — are the only things it removes, and removing them is the whole of its cost difference.

It is cheaper, and by the amount the forward pass costs: 416 s per optimization step against SIGNED's 516 at the same seed, a saving of 19.3%. Both are the window's training steps on the four-GPU box, where every arm is substantially faster than on the two-GPU seeds, so neither is comparable to a step time quoted for a different box.

It is also behind on accuracy. Over the 8 validation points of the converged window, SIGNED at seed 4 leads it by $+.0112$ on the endpoint this campaign reports and by $+.0104$ under majority vote. We therefore keep A2 and A3 rather than replacing the forecaster with the realised value, and we state the weight of that decision exactly: one run on each side, one seed. The gap is larger than SIGNED's own range across seeds, .0016, and smaller than GRPO's, .0210. It orders the two arms on this seed; it does not measure the difference between them, and the measurement that would is this arm at more seeds.

Clipping. The relaxed-clipping setting is where entropy interventions are supposed to matter most, because ratio clipping otherwise removes much of the update a token weight would have scaled. The row is one SIGNED run with no finished control under the same clipping, and it ran on a different machine from the seed-1 SIGNED reference, so it carries no contrast in either direction. It shows that the SIGNED configuration trains to the end of the schedule with the clip relaxed; it does not say whether relaxing the clip changes what the correction does.

E.11 Compute-matched comparison

An equal-step comparison is compute-fair only if the two arms cost the same per step, and here that depends on the machine. On the 4-GPU box (seeds $3, 4$) SIGNED's step costs 0.99 times GRPO's, so equal steps is equal compute to within the seed

spread of either arm (Appendix E.12). On the 2-GPU box (seed 1) it costs 1.49 times as much, and a GRPO run given the same wall clock would reach about step 164. We did not train GRPO past step 110, so that point is not measured, and we do not extrapolate a trajectory to it.

What the split does measure is where the headline gap comes from, and it is not evenly spread. SIGNED − GRPO is +.0145 on the seed where SIGNED also had the larger compute budget, and +.0011 averaged over the 2 seeds where the comparison is compute-fair. The across-seed mean of +.0055 in Table 10 is therefore carried mostly by a seed on which the treatment was also granted about half as much again in wall clock, and on the compute-fair seeds the gap is not distinguishable from zero. We report this as the reading the evidence supports: this paper does not show that SIGNED buys accuracy over GRPO at equal compute. None of this touches the contrasts that carry the structural claim — SIGNED − UNIFORM and SIGNED − PERMUTED — whose arms share the tree sampler and cost 493, 500 and 538 s per step on the larger box.

E.12 Cost

Step times are read inside a box. A seed index also names a machine (Appendix E.8), and the two boxes are further apart than any two arms are, so a mean over both measures the campaign's scheduling.

On the 4-GPU box, where every arm has finished 2 seeds, one optimization step takes 449 s for STEER, 493 for SIGNED, 500 for UNIFORM, 538 for PERMUTED and 500 for GRPO. SIGNED's overhead over STEER there is +10%. The three tree arms take 493, 500 and 538 s while STEER, which has neither the tree nor the heads, sits below all three: the cost is the rollout and the forward pass, not the A_H computation, which is a subtraction over a mask.

On the 2-GPU box the same overhead reads +86%. That is not a disagreement to average away. The tree rollout and the MTP forward are a larger share of a step when fewer GPUs serve it, so the overhead is a property of the deployment as much as of the correction, and a single figure quoted without its box — as earlier drafts of this section quoted the smaller one — reports the machine.

Neither figure is resolved to better than the spread underneath it. On the larger box STEER's 2 seeds differ by 54 s and SIGNED's by 46, against a gap between their means of the same order. On the smaller box every arm has 1 seed inside 1, 3, 4, so there is no spread to put beside it at all. We therefore report these as the cost of the runs we made and not as a coefficient of the correction. What does isolate the heads from the tree is the ORACLE arm of Appendix E.10, which removes the heads at fixed tree and fixed seed.

E.13 A note on how these numbers were selected

We report window means rather than the best checkpoint, and the difference is not cosmetic. Selecting the maximum over the twelve validation points inflates every arm, averaged over the 3 seeds: by .0161 for GRPO, .0172 for STEER, .0128 for UNIFORM, .0182 for PERMUTED and .0189 for SIGNED, relative to the window mean. Since a single AIME24 measurement carries a standard error of .0282 across problems, a maximum over a dozen noisy points is a biased statistic whose bias grows with the number of checkpoints compared, and it is not the same for every arm: the lifts differ by more than the gap between UNIFORM and PERMUTED, so ranking by the maximum would swap those two. We regard the window mean as the defensible statistic and use it throughout. The one exception is Appendix F, which uses the maximum on purpose, because that is the rule of the table it is compared against.

F Calibration Against the Reported Numbers

Every number before this section is our own measurement under the schedule of Section 5, which differs from the base method's in length, checkpoint rule and number of runs (Appendix E.10). This section is the one place a figure is taken from prior work. It runs the SIGNED configuration under the base method's own protocol and sets the result beside that method's published 1.5B table (Hao et al., 2026, Table 12), which also reports GRPO and the untrained model. It is a second kind of evidence, not a sixth arm, and we say below what it cannot show.

Protocol. Two independent runs, with data seeds 1 and 2, of 200 optimization steps each; AIME24 validation every 10 steps; in each run the checkpoint with the highest AIME24 accuracy is selected and evaluated on the six benchmarks of Appendix E.9, and a table entry is the mean over the

two runs. Each of these items is the base method's (Hao et al., 2026, App. E). Everything else is the configuration of Section 5, with one change: the tree sampler uses two roots, each cut at depths $\{64, 192\}$ with branching factor 2, so the eight responses to a prompt form two independent groups of four instead of one group that shares its first 64 tokens. The purpose of the change was to keep the siblings the estimator needs while sharing less of each response, because short runs after the five-arm campaign showed the single-root tree making the eight rewards of a group tie more often, which removes signal from a GRPO update. It is an adjustment to keep the estimator's support affordable, not an attempt to improve the sampler. The choice is exploratory, it was made on training prompts and not on any benchmark, and these runs are not one of the five arms. The selected steps are 150 and 190.

Each AIME24 cell of ours is read from the training log and used only if it agrees with the reward printed beside it and with the trainer's own record of the best value; the other columns pass the coherence check of Appendix E.9 (`scripts/emit_r2_numbers.py`). The two-run AIME24 mean is averaged over correct-sample counts rather than over two rounded fractions.

Reading the table. Averaged over the two runs, the SIGNED arm's six-benchmark mean is 36.7, against 38.2 for the base method and 35.8 for GRPO as reported; on AIME24 the same three figures are 17.4, 17.4 and 16.2. One AIME24 evaluation carries an across-problem standard error of .0282 as a fraction (Appendix D.2); a mean of two runs halves its variance and no more, and the quoted rows carry a noise of their own that their source does not state. Against the base method's reported row the two-run mean is lower on 5 of the six columns and not lower on AIME24, the column the selection itself used; against the reported GRPO row it is higher on 5. We therefore read the table as placing the SIGNED arm between the two reported rows and below the base method as reported, not level with it; how much of that gap belongs to the correction and how much to the environment, an uncontrolled comparison cannot say. The step-200 row is printed because the selected checkpoint is chosen on AIME24 and so flatters its own row; the last checkpoint is chosen by nobody.

What this comparison is not. It is not controlled. The quoted rows ran with the authors' code on different hardware (a cluster of H20 GPUs, against our four H100s), and the base method under this protocol in our environment is a run this paper does not have. The controlled comparison with the base method is the five-arm campaign, where SIGNED − STEER is +.0120 over 3 paired seeds at 110 steps (Table 10), with the caveats of Appendix E.10. Nor does the table speak to the visitation channel: like every comparison between the full and the base method (Section 5), it changes the sampler and the correction together, and the attribution rests on the controls of Section 6 alone.

G Closed Forms for Section 3.1

Perturbing one logit of a k-way uniform distribution by δ and writing $S := e^{\delta} + k - 1$ gives $\pi_1 = e^{\delta}/S$ and $\pi_j = 1/S$ for $j \neq 1$, so

$$H(\delta) = \log S - \frac{\delta e^{\delta}}{S},$$
$$H'(\delta) = -\frac{(k-1)\,\delta\, e^{\delta}}{S^2},$$

whence $H'(0) = 0$ and $H''(0) = -(k-1)/k^2$. The sign of H' is $-\operatorname{sign}(\delta)$, so any perturbation lowers the entropy, but only at second order. Table 12 contrasts this with a peaked distribution, where the first-order term does not vanish.

distribution	H_0	Ω	ΔH at δ=.1
uniform, k=10	2.303	0	-4.7×10^{-4}
uniform, k=50	3.912	0	-1.0×10^{-4}
top .5 + 9 uniform	1.792	−0.549	-5.6×10^{-2}
top .9 + 9 uniform	0.545	−0.396	-3.8×10^{-2}

Table 12: First-order prediction versus actual entropy change. At a uniform branch point the first-order term is exactly zero and the realized damage is two orders of magnitude below that of a peaked state under the same perturbation.

H Implementation Notes

Sibling mask. For a group of G rollouts of length T we form $\mathrm{eq}[i, j, t] = \mathbf{1}[y_t^{(i)} = y_t^{(j)}]$ and take a cumulative product along t, so that $\mathrm{prefixeq}[i, j, t]$ is true exactly when the two rollouts agree on $y_{\le t}$. Shifting it one position right gives agreement on $y_{<t}$, which is the sibling relation Eq. (5) requires. Padded positions are treated as disagreement, so a length difference counts as a divergence.

Alignment. $\hat{H}_{\text{togo}}$ is read at position $t+1$ and moved to position t; at the final position the value is kept. Omitting this shift scores the state before the branch instead of the branch taken.

Memory. The RL path never materializes head logits. For each head the hidden states are processed in chunks of 4096 rows, mapped through the shared unembedding, reduced to an entropy, and released, so peak memory is $O(\text{chunk} \times V)$ rather than $O(\kappa \times B \times T \times V)$.

Equivalence test. A unit test asserts that with $\lambda = 0$ the token weights are bitwise identical to the base method's on random inputs; the $\lambda = 0$ early return in the combination step is what makes this hold exactly rather than to floating-point tolerance.

I Pre-Registered Analysis

The analysis plan below was committed, with a timestamp, to the project repository (to be made public with the code) before any run of the multi-seed campaign was launched, so that the statistics could not be chosen to fit the outcome. It is reproduced here in full; the file is `docs/preregistration_5seed.md` in that repository. Arm names follow the main text; the registration's wording is otherwise unchanged.

Unit and primary endpoint. The unit of analysis is the seed, not the validation step. Each run is reduced to one number — its mean acc@32 on AIME24 over steps 40–110 — and arms are paired within a seed.

Contrasts, in the order they are reported. Primary: SIGNED − GRPO. Secondary, in this order: SIGNED − STEER, SIGNED − UNIFORM, SIGNED − PERMUTED. Secondary endpoints: maj@32, uplift, and the six-benchmark mean. Effect sizes are reported with intervals rather than significance stars, and all five arms are reported whatever the outcome.

Stopping and exclusion. Every run goes to step 110; none is stopped early for looking good or bad. A run is excluded only for mechanical failure — out of memory, a crash, a corrupted checkpoint, or a step-0 validation score outside .030–.055, which would indicate a different initial checkpoint. Exclusions are listed with their reason. A failed run is relaunched with the same seed, never a new one, and no seed is added after results are seen.

Checkpoint selection. Two rules are computed and both reported: the argmax over a run's validation points, and a fixed step 110. The six-benchmark evaluation uses the argmax. Selecting a maximum over twelve noisy checkpoints is biased upward — in the single-seed data by $+.011$ to $+.017$ per arm — which is why the primary endpoint is a window mean rather than a maximum.

What would falsify the central claim. Stated in advance so it cannot be renegotiated afterwards. If SIGNED − UNIFORM straddles zero across seeds, the forecast's *value* does not matter and the apparatus explains the effect; that is the claim of this paper and it would be refuted. If SIGNED − PERMUTED straddles zero, the pairing between a branch and its score does not matter, which the derivation says it must. If aggregate entropy orders the arms the same way accuracy does, the dissociation argument of Section 6.1 fails, since it depends on the two orderings coming apart. And if the mean token weight in the SIGNED arm departs from ≈ 0.999, then A_H is not sum-zero over sibling sets as derived and the implementation is wrong rather than the theory.

Amendment, recorded. The pre-registration assigned three seeds to UNIFORM and PERMUTED and five to the other three arms. The campaign as executed runs all five arms at five seeds. The change raises the power of the two control contrasts and was made before any result of the campaign was seen; it is recorded here rather than silently absorbed.

Deviations as executed. Recorded here because the plan above was not carried out in full. (i) Seeds: every arm completed seeds 1–4 except SIGNED at seed 2, which failed at start-up and was not relaunched; the seed-5 runs were started and not completed. The analysis therefore uses the seeds on which all five arms finished, $1, 3, 4$, and the other runs are in the supplementary logs but enter no statistic. (ii) Machines: seed 1 ran on a two-GPU box and seeds 3 and 4 on a four-GPU box, for every arm (Appendix E.8). (iii) The six-benchmark evaluation covers one seed per arm and no STEER checkpoint (Appendix E.9). (iv) The long GRPO run for the compute-matched comparison was not completed; Appendix E.11 reports what the step costs support instead. (v) The planned second backbones were not run. (vi) The support-restricted split registered below is instrumented but carried by no run reported here. (vii) Follow-up arms: the four

advantage-estimator runs (RLOO and OPO, each with SIGNED and STEER) failed at start-up on infrastructure errors and completed no training step; the base method at the released $\lambda_{\min} = 0.8$ was not launched; the relaxed-clipping STEER control stopped at step 62. None of them enters Table 7, and their logs are in the supplementary material. The choice to analyse on the common seed set was made after per-seed results were visible; it is the only set on which all five arms exist, and every run outside it is in the supplementary logs.

Second amendment, added 2026-09-21, after this section's own entropy condition was retired. Everything above was fixed before the campaign ran. This paragraph was not, and it is marked so rather than folded into the text above. It justifies nothing about the seeds already collected; it binds the seeds not yet in hand.

What it replaces. The third falsification bullet above put a condition on the aggregate entropy ordering. Section 6.1 retires it as a category error — it registers the aggregate as a signal about A_H, which Section 4 had already said it cannot be, an hour before this file was committed — and claims nothing from it in either direction. A first attempt at a replacement, registered earlier on 2026-09-21, named `steerf/branch_entropy`; that registration is withdrawn here rather than deleted, because the monitor it named selects a fixed top decile of A_H rather than the support (Appendix E.6).

Registered now, before the runs that can answer it. The quantity is `steerf/support_entropy` and its gap, the entropy split on $A_H \neq 0$ — the same predicate the correction itself applies, so the split covers the positions the correction touches and no others. The primary contrast is SIGNED − PERMUTED, paired by seed, because that pair shares λ, $\lambda_{\min}$, the sampler and the magnitudes of A_H, and differs only in whether a branch is paired with its own score; SIGNED − UNIFORM is secondary. The claim under test is that a support-restricted entropy contrast near zero can accompany an accuracy contrast away from zero.

What would falsify it. If the support-restricted contrast tracks the accuracy contrast in sign and magnitude across seeds the way the apparatus contrasts in Table 4 do, then the split is the aggregate statistic in a smaller window and the claim fails.

What is still not instrumented, and we say so rather than substitute. Section 4's own prediction is that the spread *across siblings* widens. Testing it needs the sibling index at the loss site, where only the per-token tensors are currently in scope, so no quantity we log tests it today. We do not offer the support-restricted split as that test: it measures a level, not a spread.

J Within-Run Stability

Earlier drafts of this work paired the arms at each of the eight validation steps in the converged window and reported a paired t over those eight points. That statistic answers a narrower question than the one in Section 6.3: it asks whether a gap is stable along a single trajectory, not whether it reappears when the run is repeated. Its n counts validation points, which are serially correlated within a run, so it is not an estimate of run-to-run uncertainty and cannot be read as one.

We report it here because it is informative about a different thing — whether a gap is an artifact of one lucky validation point — and because the single-seed version of this manuscript quoted it. Where the two disagree, the seed-level statistic in Appendix E.8 governs.

The contrast between the two is the point. Table 13 separates every A_H comparison at t_7 between $+2.56$ and $+7.18$, which reads as decisive and is not: the same gap sits at the same order as the run-to-run spread we measure in Appendix E.8. A statistic can be this confident about a trajectory and say nothing about a re-run, which is exactly why we moved it out of the main text.

contrast	Δacc@32	paired t_7
SIGNED − GRPO	+.0145	+3.08
SIGNED − STEER	+.0165	+6.68
SIGNED − UNIFORM	+.0182	+7.18
SIGNED − PERMUTED	+.0125	+2.56

Table 13: Within-run stability over the eight converged-window validation points of seed 1. The points are consecutive validations of one run and are therefore serially correlated, so $\mathrm{df} = 7$ is generous and these t values are not an estimate of run-to-run variation; see Appendix E.8 for that. Majority-vote versions of the same four rows separate at $+5.09$ to $+5.50$.